

# Sid Mathur's Thematic ETF Screener – Daily Report

2026-07-15

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## Action Center

**Run date:** 2026-07-15   **Universe:** 660 ETFs / 26 clusters

**Full PC1:** FLAT   **Clusters long:** 6/26   **ETFs long:** 180/660 (27%)

## Today's events (cluster level)

spread\_buy = 5dma – 50dma; spread\_sell = 5dma – 20dma (price units). A BUY fired today iff spread\_buy crossed above 0; a SELL iff spread\_sell crossed below the cushioned threshold.

### SELL (1)

| cluster   | theme               | spread_sell | p_h5 | nss   |
|-----------|---------------------|-------------|------|-------|
| cluster_1 | U.S. Mid Cap Equity | -0.545      | 0.9  | 0.096 |

## Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p\_h5 is the 5-day probability for context. **spread\_buy = 5dma – 50dma, spread\_sell = 5dma – 20dma** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

### Imminent BUY (top 5 by h=1 NSS, filter-approved)

| cluster    | theme                       | spread_buy | p_h1  | p_h5 | nss_h1 | nss_h5 |
|------------|-----------------------------|------------|-------|------|--------|--------|
| cluster_14 | Financials                  | 6.176      | 1.000 | 1.00 | 1.000  | 1.00   |
| cluster_5  | U.S. Large Cap Equity       | 1.546      | 1.000 | 0.89 | 0.993  | 0.84   |
| cluster_15 | Real Estate                 | 1.442      | 1.000 | 0.91 | 0.797  | 0.43   |
| cluster_22 | Japan Equity                | 2.020      | 0.984 | 0.81 | 0.230  | 0.21   |
| cluster_7  | U.S. Extended Market Equity | 0.783      | 0.934 | 0.65 | 0.103  | 0.05   |

### Imminent SELL (top 3 by h=1 NSS)

| cluster    | theme                      | spread_sell | p_h1 | p_h5 | nss_h1 | nss_h5 |
|------------|----------------------------|-------------|------|------|--------|--------|
| cluster_25 | International Fixed Income | -0.260      | 0.99 | 0.81 | -0.990 | -0.81  |
| cluster_3  | Developed ex-US Equity     | -0.414      | 0.84 | 0.65 | -0.715 | -0.44  |
| cluster_19 | Broad Commodities          | 3.587       | 0.00 | 0.01 | 0.000  | 0.00   |

## Full Sample PC1



Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

**Replication basket:** VOO basket cor: 0.94 sticky

**Current state:** FLAT

5dma: 142.136 20dma: 142.268 50dma: 141.311

Spread vs 50dma: +0.825 Spread vs 20dma: -0.132

P(buy crossover within 5 days): 79.9% P(sell): 54.7%

## Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

### Active positions (long clusters)

| ticker | cluster    | theme                      | nss    | days | return | pos_wt | cor  |
|--------|------------|----------------------------|--------|------|--------|--------|------|
| VHT    | cluster_11 | Healthcare                 | 0.992  | 46   | 9.8%   | 12.86% | 0.93 |
| VTV    | cluster_2  | U.S. Large Cap Value       | 0.975  | 63   | 6.4%   | 20.01% | 0.98 |
| XLP    | cluster_18 | Consumer Staples           | 0.663  | 11   | -0.9%  | 8.84%  | 0.99 |
| XLU    | cluster_18 | Consumer Staples           | 0.663  | 11   | -0.9%  | 5.47%  | 0.99 |
| PDBC   | cluster_19 | Broad Commodities          | -0.004 | 27   | -7.1%  | 8.37%  | 0.98 |
| VEA    | cluster_3  | Developed ex-US Equity     | -0.439 | 6    | -1.3%  | 11.37% | 0.99 |
| BWX    | cluster_25 | International Fixed Income | -0.813 | 30   | -2.3%  | 15.01% | 0.93 |

### Watchlist (imminent-buy clusters, $p > 80\%$ , not yet long)

| ticker | cluster    | theme                 | p_h5 | nss   | weight | cor  |
|--------|------------|-----------------------|------|-------|--------|------|
| XLF    | cluster_14 | Financials            | 1.00 | 0.999 | 1.095  | 0.96 |
| IJH    | cluster_1  | U.S. Mid Cap Equity   | 0.99 | 0.096 | 0.982  | 0.99 |
| VNQ    | cluster_15 | Real Estate           | 0.91 | 0.433 | 0.970  | 0.99 |
| VOO    | cluster_5  | U.S. Large Cap Equity | 0.89 | 0.839 | 0.936  | 1.00 |
| EWJ    | cluster_22 | Japan Equity          | 0.81 | 0.212 | 0.910  | 0.97 |

## Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

|    | id         | name                           | category             | n  | signal | change  | days | return | nss    | dNSS   |
|----|------------|--------------------------------|----------------------|----|--------|---------|------|--------|--------|--------|
| 14 | cluster_14 | Financials                     | Financials           | 14 | FLAT   |         |      |        | 0.999  | -0.001 |
| 11 | cluster_11 | Healthcare                     | Healthcare           | 17 | LONG   |         | 46   | 9.8%   | 0.992  | -0.008 |
| 2  | cluster_2  | U.S. Large Cap Value           | U.S. Large Cap Va... | 68 | LONG   |         | 63   | 6.4%   | 0.975  | -0.017 |
| 5  | cluster_5  | U.S. Large Cap Equity          | U.S. Large Cap Eq... | 47 | FLAT   |         |      |        | 0.839  | +0.008 |
| 18 | cluster_18 | Consumer Staples               | Consumer Staples     | 11 | LONG   |         | 11   | -0.9%  | 0.663  | -0.170 |
| 9  | cluster_9  | Asia-Pacific Equity            | Asia-Pacific Equi... | 24 | FLAT   |         |      |        | 0.476  | +0.134 |
| 15 | cluster_15 | Real Estate                    | Real Estate          | 14 | FLAT   |         |      |        | 0.433  | -0.203 |
| 22 | cluster_22 | Japan Equity                   | Japan Equity         | 9  | FLAT   |         |      |        | 0.212  | +0.010 |
| 1  | cluster_1  | U.S. Mid Cap Equity            | U.S. Mid Cap Equi... | 91 | FLAT   |         |      |        | 0.096  | -0.100 |
| 7  | cluster_7  | U.S. Extended Market Equity    | U.S. Extended Mar... | 42 | FLAT   |         |      |        | 0.051  | -0.091 |
| 19 | cluster_19 | Broad Commodities              | Broad Commodities    | 11 | LONG   |         | 27   | -7.1%  | -0.004 | +0.040 |
| 13 | cluster_13 | China Equity                   | China Equity         | 15 | FLAT   |         |      |        | -0.040 | +0.068 |
| 26 | cluster_26 | Leveraged / Inverse            | Leveraged / Inver... | 2  | FLAT   | shifted |      |        | -0.258 | -0.106 |
| 20 | cluster_20 | India Equity                   | India Equity         | 10 | FLAT   |         |      |        | -0.403 | -0.119 |
| 3  | cluster_3  | Developed ex-US Equity         | Developed ex-US E... | 61 | LONG   |         | 6    | -1.3%  | -0.439 | -0.146 |
| 16 | cluster_16 | Municipal Bond                 | Municipal Bond       | 13 | FLAT   |         |      |        | -0.557 | -0.185 |
| 12 | cluster_12 | Materials                      | Materials            | 17 | FLAT   |         |      |        | -0.708 | +0.184 |
| 6  | cluster_6  | U.S. Large Cap Growth          | U.S. Large Cap Gr... | 44 | FLAT   |         |      |        | -0.800 | +0.067 |
| 17 | cluster_17 | Gold                           | Gold                 | 12 | FLAT   |         |      |        | -0.804 | -0.014 |
| 25 | cluster_25 | International Fixed Income     | International Fix... | 6  | LONG   |         | 30   | -2.3%  | -0.813 | +0.053 |
| 8  | cluster_8  | Emerging Markets Equity        | Emerging Markets ... | 25 | FLAT   |         |      |        | -0.906 | -0.013 |
| 24 | cluster_24 | Cash / Ultra-Short Fixed In... | Cash / Ultra-Shor... | 9  | FLAT   |         |      |        | -0.906 | +0.028 |
| 23 | cluster_23 | Bank Loan / Floating Rate      | Bank Loan / Float... | 9  | FLAT   |         |      |        | -0.992 | +0.000 |
| 10 | cluster_10 | High Yield Bond                | High Yield Bond      | 21 | FLAT   |         |      |        | -0.997 | -0.007 |
| 21 | cluster_21 | Inflation-Protected Fixed I... | Inflation-Protect... | 10 | FLAT   |         |      |        | -0.997 | +0.001 |
| 4  | cluster_4  | Aggregate Fixed Income         | Aggregate Fixed I... | 58 | FLAT   |         |      |        | -0.998 | -0.003 |

## Cluster Details

### Cluster 1 – U.S. Mid Cap Equity

**Members:** 91   **Var explained:** 0.854   **Signal:** FLAT

#### Signal logic:

- Latest buy\_cross: 2026-04-13 – filter **PASS** (252d at date = +30.0%)
- Latest sell\_cross: 2026-07-14
- Current 252d return: +19.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-14 more recent than last effective buy 2026-04-13)



#### Category breakdown

| category               | n_members | abs_weight |
|------------------------|-----------|------------|
| Industrials            | 14        | 0.1450     |
| U.S. Small Cap Equity  | 10        | 0.1161     |
| Consumer Discretionary | 11        | 0.1111     |
| U.S. Mid Cap Equity    | 6         | 0.0703     |
| U.S. Small Cap Value   | 6         | 0.0692     |
| Materials              | 4         | 0.0409     |
| U.S. Mid Cap Value     | 3         | 0.0351     |
| U.S. Small Cap Growth  | 3         | 0.0348     |

|                                       |   |        |
|---------------------------------------|---|--------|
| U.S. Mid Cap Growth                   | 3 | 0.0343 |
| Utilities                             | 3 | 0.0311 |
| Dividend Equity                       | 2 | 0.0223 |
| Leveraged / Inverse                   | 2 | 0.0223 |
| U.S. Small Cap Low Volatility         | 2 | 0.0212 |
| Unknown                               | 1 | 0.0118 |
| U.S. Small Cap Fundamental Equity     | 1 | 0.0118 |
| U.S. Mid/Small Cap Equity             | 1 | 0.0117 |
| U.S. Small Cap ESG Equity             | 1 | 0.0117 |
| Mid Cap Value Equities                | 1 | 0.0117 |
| U.S. Small/Mid Cap Fundamental Equity | 1 | 0.0117 |
| U.S. Small Cap Multifactor Equity     | 1 | 0.0116 |
| U.S. Mid Cap Dividend Equity          | 1 | 0.0115 |
| Small Cap Value Equities              | 1 | 0.0114 |
| U.S. ESG Equity                       | 1 | 0.0114 |
| U.S. Small/Mid Cap Equity             | 1 | 0.0114 |
| U.S. Mid Cap Quality Equity           | 1 | 0.0113 |
| Value Equity                          | 1 | 0.0112 |
| U.S. Small Cap Dividend Equity        | 1 | 0.0112 |
| U.S. Equity                           | 1 | 0.0112 |
| U.S. Small Cap Industrials            | 1 | 0.0111 |
| U.S. Small Cap Momentum Equity        | 1 | 0.0110 |
| U.S. Micro Cap Equity                 | 1 | 0.0107 |
| U.S. Small Cap Consumer Discretionary | 1 | 0.0106 |
| U.S. Dividend Equity                  | 1 | 0.0106 |
| U.S. Mid Cap Low Volatility           | 1 | 0.0102 |
| Real Estate                           | 1 | 0.0091 |

**Replication basket** – IJH (cor=0.99) basket lead at lag 1: -0.008

| ticker | rep_wt | name                         | category            | etf_signal |
|--------|--------|------------------------------|---------------------|------------|
| IJH    | 0.982  | iShares Core S&P Mid-Cap ETF | U.S. Mid Cap Equity | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category              | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------|------------|
| 30 | MDY    | 0.0118 | State Street SPDR S&P MIDCAP 400 ETF Tr... | U.S. Mid Cap Equity   | FLAT       |
| 37 | IVOO   | 0.0118 | Vanguard S&P Mid-Cap 400 ETF               |                       | FLAT       |
| 41 | IJH    | 0.0118 | iShares Core S&P Mid-Cap ETF               | U.S. Mid Cap Equity   | FLAT       |
| 42 | SPMD   | 0.0118 | State Street SPDR Portfolio S&P 400 Mid... | U.S. Mid Cap Equity   | FLAT       |
| 44 | VB     | 0.0118 | Vanguard Small-Cap ETF                     | U.S. Small Cap Equity | FLAT       |

|    |       |        |                                            |                             |      |
|----|-------|--------|--------------------------------------------|-----------------------------|------|
| 63 | FNDA  | 0.0118 | Schwab Fundamental U.S. Small Company E... | U.S. Small Cap Fundament... | FLAT |
| 75 | VBR   | 0.0118 | Vanguard Small-Cap Value ETF               | U.S. Small Cap Value        | FLAT |
| 17 | JHMM  | 0.0117 | John Hancock Multifactor Mid Cap ETF       | U.S. Mid Cap Equity         | FLAT |
| 23 | SCHM  | 0.0117 | Schwab U.S. Mid Cap ETF                    | U.S. Mid Cap Equity         | FLAT |
| 24 | FNX   | 0.0117 | First Trust Mid Cap Core AlphaDEX Fund     | U.S. Mid/Small Cap Equity   | FLAT |
| 27 | NUSC  | 0.0117 | Nuveen ESG Small-Cap ETF                   | U.S. Small Cap ESG Equity   | FLAT |
| 28 | PRFZ  | 0.0117 | Invesco RAFI US 1500 Small-Mid ETF         | U.S. Small/Mid Cap Funda... | FLAT |
| 38 | VIOO  | 0.0117 | Vanguard S&P Small-Cap 600 ETF             | U.S. Small Cap Equity       | FLAT |
| 40 | IJR   | 0.0117 | iShares Core S&P Small-Cap ETF             | U.S. Small Cap Equity       | FLAT |
| 43 | SPSM  | 0.0117 | State Street SPDR Portfolio S&P 600 Sma... | U.S. Small Cap Equity       | FLAT |
| 58 | EZM   | 0.0117 | WisdomTree U.S. MidCap Fund                | Mid Cap Value Equities      | FLAT |
| 68 | IJJ   | 0.0117 | iShares S&P Mid-Cap 400 Value ETF          | U.S. Mid Cap Value          | FLAT |
| 70 | IVOV  | 0.0117 | Vanguard S&P Mid-Cap 400 Value ETF         | U.S. Mid Cap Value          | FLAT |
| 71 | MDYV  | 0.0117 | State Street SPDR S&P 400 Mid Cap Value... | U.S. Mid Cap Value          | FLAT |
| 7  | IWM   | 0.0116 | iShares Russell 2000 Index Fund            | U.S. Small Cap Equity       | FLAT |
| 25 | FYX   | 0.0116 | First Trust Small Cap Core AlphaDEX Fund   | U.S. Small Cap Equity       | FLAT |
| 29 | SMLF  | 0.0116 | iShares U.S. Small-Cap Equity Factor ETF   | U.S. Small Cap Multifact... | FLAT |
| 31 | IJT   | 0.0116 | iShares S&P SmallCap 600 Growth ETF        | U.S. Small Cap Growth       | FLAT |
| 33 | VIOG  | 0.0116 | Vanguard S&P Small-Cap 600 Growth ETF      | U.S. Small Cap Growth       | FLAT |
| 36 | SLYG  | 0.0116 | State Street SPDR S&P 600 Small Cap Gro... | U.S. Small Cap Growth       | FLAT |
| 45 | SCHA  | 0.0116 | Schwab U.S. Small-Cap ETF                  | U.S. Small Cap Equity       | FLAT |
| 66 | IWN   | 0.0116 | iShares Russell 2000 Value ETF             | U.S. Small Cap Value        | FLAT |
| 83 | RWK   | 0.0116 | Invesco S&P MidCap 400 Revenue ETF         | U.S. Mid Cap Equity         | FLAT |
| 11 | FXR   | 0.0115 | First Trust Industrials AlphaDEX           | Industrials                 | FLAT |
| 39 | VTWO  | 0.0115 | Vanguard Russell 2000 ETF                  | U.S. Small Cap Equity       | FLAT |
| 56 | DON   | 0.0115 | WisdomTree U.S. MidCap Dividend Fund       | U.S. Mid Cap Dividend Eq... | FLAT |
| 59 | PSC   | 0.0115 | Principal U.S. Small-Cap ETF               | U.S. Small Cap Equity       | FLAT |
| 69 | IJS   | 0.0115 | iShares S&P SmallCap 600 Value ETF         | U.S. Small Cap Value        | LONG |
| 72 | SLYV  | 0.0115 | State Street SPDR S&P 600 Small Cap Val... | U.S. Small Cap Value        | FLAT |
| 74 | VTWV  | 0.0115 | Vanguard Russell 2000 Value ETF            | U.S. Small Cap Value        | FLAT |
| 90 | SDVY  | 0.0115 | First Trust SMID Cap Rising Dividend Ac... | Dividend Equity             | LONG |
| 32 | IJK   | 0.0114 | iShares S&P Mid-Cap 400 Growth ETF         | U.S. Mid Cap Growth         | FLAT |
| 34 | IVOG  | 0.0114 | Vanguard S&P Mid-Cap 400 Growth ETF        | U.S. Mid Cap Growth         | FLAT |
| 35 | MDYG  | 0.0114 | State Street SPDR S&P 400 Mid Cap Growt... | U.S. Mid Cap Growth         | FLAT |
| 57 | EES   | 0.0114 | WisdomTree U.S. SmallCap Fund              | Small Cap Value Equities    | FLAT |
| 73 | VIOV  | 0.0114 | Vanguard S&P Small-Cap 600 Value ETF       | U.S. Small Cap Value        | FLAT |
| 76 | ETHO  | 0.0114 | Amplify Etho Climate Leadership U.S. ETF   | U.S. ESG Equity             | FLAT |
| 77 | RWJ   | 0.0114 | Invesco S&P SmallCap 600 Revenue ETF       | U.S. Small Cap Equity       | FLAT |
| 91 | ISMID | 0.0114 | Inspire Small/Mid Cap ETF                  | U.S. Small/Mid Cap Equity   | LONG |
| 80 | XMHQ  | 0.0113 | Invesco S&P MidCap Quality ETF             | U.S. Mid Cap Quality Equ... | FLAT |
| 82 | RFV   | 0.0113 | Invesco S&P MidCap 400 Pure Value ETF      | Leveraged / Inverse         | FLAT |
| 55 | DES   | 0.0112 | WisdomTree U.S. SmallCap Dividend Fund     | U.S. Small Cap Dividend ... | FLAT |
| 84 | OMFS  | 0.0112 | Invesco Russell 2000 Dynamic Multifacto... | U.S. Equity                 | FLAT |
| 89 | FYT   | 0.0112 | First Trust Small Cap Value AlphaDEX Fu... | Value Equity                | FLAT |
| 15 | IYJ   | 0.0111 | iShares U.S. Industrials ETF               | Industrials                 | FLAT |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 21 | VIS  | 0.0111 | Vanguard Industrials ETF                   | Industrials                 | FLAT |
| 22 | FIDU | 0.0111 | Fidelity MSCI Industrials Index ETF        | Industrials                 | FLAT |
| 46 | FXD  | 0.0111 | First Trust Cons. Discret. AlphaDEX        | Consumer Discretionary      | FLAT |
| 78 | PSCI | 0.0111 | Invesco S&P SmallCap Industrials ETF       | U.S. Small Cap Industria... | FLAT |
| 79 | PAVE | 0.0111 | Global X U.S. Infrastructure Developmen... | Industrials                 | FLAT |
| 85 | RZV  | 0.0110 | Invesco S&P SmallCap 600 Pure Value ETF    | Leveraged / Inverse         | FLAT |
| 88 | XSMO | 0.0110 | Invesco S&P SmallCap Momentum ETF          | U.S. Small Cap Momentum ... | FLAT |
| 53 | RSPD | 0.0109 | Invesco S&P 500 Equal Weight Consumer D... | Consumer Discretionary      | FLAT |
| 2  | XLI  | 0.0108 | State Street Industrial Select Sector S... | Industrials                 | FLAT |
| 14 | FIW  | 0.0108 | First Trust Water ETF                      | Utilities                   | LONG |
| 18 | REGL | 0.0108 | ProShares S&P MidCap 400 Dividend Arist... | Dividend Equity             | LONG |
| 48 | FTXR | 0.0108 | First Trust Nasdaq Transportation ETF      | Industrials                 | LONG |
| 67 | SMMV | 0.0108 | iShares MSCI USA Small-Cap Min Vol Fact... | U.S. Small Cap Low Volat... | LONG |
| 26 | IWC  | 0.0107 | iShares Microcap ETF                       | U.S. Micro Cap Equity       | FLAT |
| 54 | SMDV | 0.0106 | ProShares Russell 2000 Dividend Growers... | U.S. Dividend Equity        | FLAT |
| 87 | PSCD | 0.0106 | Invesco S&P SmallCap Consumer Discretio... | U.S. Small Cap Consumer ... | LONG |
| 12 | PHO  | 0.0105 | Invesco Water Resources ETF                | Utilities                   | LONG |
| 20 | VAW  | 0.0105 | Vanguard Materials ETF                     | Materials                   | FLAT |
| 62 | XTN  | 0.0105 | State Street SPDR S&P Transportation ETF   | Industrials                 | FLAT |
| 65 | XSLV | 0.0104 | Invesco S&P SmallCap Low Volatility ETF    | U.S. Small Cap Low Volat... | FLAT |
| 81 | PKB  | 0.0104 | Invesco Building & Construction ETF        | Industrials                 | FLAT |
| 8  | IYC  | 0.0103 | iShares U.S. Consumer Discretionary ETF    | Consumer Discretionary      | LONG |
| 19 | RSPM | 0.0103 | Invesco S&P 500 Equal Weight Materials ... | Materials                   | LONG |
| 3  | XLB  | 0.0102 | State Street Materials Select Sector SP... | Materials                   | FLAT |
| 4  | XRT  | 0.0102 | State Street SPDR S&P Retail ETF           | Consumer Discretionary      | LONG |
| 9  | VCR  | 0.0102 | Vanguard Consumer Discretion ETF           | Consumer Discretionary      | LONG |
| 10 | FDIS | 0.0102 | Fidelity MSCI Consumer Discretionary In... | Consumer Discretionary      | LONG |
| 52 | IYT  | 0.0102 | iShares U.S. Transportation ETF            | Industrials                 | FLAT |
| 64 | XMLV | 0.0102 | Invesco S&P MidCap Low Volatility ETF      | U.S. Mid Cap Low Volatil... | LONG |
| 61 | XHB  | 0.0101 | State Street SPDR S&P Homebuilders ETF     | Consumer Discretionary      | FLAT |
| 47 | PEJ  | 0.0100 | Invesco Leisure and Entertainment ETF      | Consumer Discretionary      | LONG |
| 16 | IYM  | 0.0099 | iShares U.S. Basic Materials ETF           | Materials                   | FLAT |
| 1  | XLY  | 0.0098 | State Street Consumer Discretionary Sel... | Consumer Discretionary      | LONG |
| 13 | CGW  | 0.0098 | Invesco S&P Global Water Index ETF         | Utilities                   | LONG |
| 49 | PPA  | 0.0093 | Invesco Aerospace & Defense ETF            | Industrials                 | FLAT |
| 60 | XAR  | 0.0093 | State Street SPDR S&P Aerospace & Defen... | Industrials                 | FLAT |
| 5  | JETS | 0.0092 | U.S. Global Jets ETF                       | Industrials                 | FLAT |
| 51 | ITB  | 0.0092 | iShares U.S. Home Construction ETF         | Consumer Discretionary      | FLAT |
| 86 | RTH  | 0.0092 | VanEck Retail ETF                          | Consumer Discretionary      | FLAT |
| 6  | REM  | 0.0091 | iShares Mortgage Real Estate ETF           | Real Estate                 | LONG |
| 50 | ITA  | 0.0087 | iShares U.S. Aerospace & Defense ETF       | Industrials                 | FLAT |

**Cluster 2 – U.S. Large Cap Value**

**Members:** 68   **Var explained:** 0.888   **Signal:** LONG since 2026-04-13 (63 days, 6.4%)

**Signal logic:**

- Latest buy\_cross: 2026-04-13 – filter **PASS** (252d at date = +23.1%)
- Latest sell\_cross: 2026-03-06
- Current 252d return: +19.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)

**Category breakdown**

| category                   | n_members | abs_weight |
|----------------------------|-----------|------------|
| U.S. Dividend Equity       | 21        | 0.3062     |
| U.S. Large Cap Value       | 10        | 0.1536     |
| Value Equity               | 4         | 0.0594     |
| U.S. Low Volatility Equity | 4         | 0.0547     |
| U.S. Multifactor Equity    | 3         | 0.0449     |
| U.S. Mid Cap Equity        | 3         | 0.0441     |
| U.S. Value Factor Equity   | 3         | 0.0438     |
| U.S. Fundamental Equity    | 2         | 0.0306     |
| U.S. Mid Cap Value         | 2         | 0.0304     |
| U.S. Equal Weight Equity   | 2         | 0.0303     |
| U.S. Equity                | 2         | 0.0303     |

|                              |   |        |
|------------------------------|---|--------|
| Dividend Equity              | 2 | 0.0282 |
| U.S. Mega Cap Value          | 1 | 0.0152 |
| U.S. Revenue Weighted Equity | 1 | 0.0151 |
| Large Cap Blend Equities     | 1 | 0.0149 |
| U.S. Size Factor Equity      | 1 | 0.0148 |
| U.S. Buyback Equity          | 1 | 0.0147 |
| U.S. Large Cap Equity        | 1 | 0.0146 |
| U.S. Quality / Factor Equity | 1 | 0.0143 |
| U.S. Defensive Equity        | 1 | 0.0140 |
| Global Equity                | 1 | 0.0135 |
| Communication Services       | 1 | 0.0122 |

**Replication basket** – VTV (cor=0.98) basket lead at lag 1: 0.006

| ticker | rep_wt | name               | category             | etf_signal |
|--------|--------|--------------------|----------------------|------------|
| VTV    | 1.001  | Vanguard Value ETF | U.S. Large Cap Value | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                 | etf_signal |
|----|--------|--------|--------------------------------------------|--------------------------|------------|
| 24 | IWD    | 0.0154 | iShares Russell 1000 Value ETF             | U.S. Large Cap Value     | LONG       |
| 27 | JPUS   | 0.0154 | JPMorgan Diversified Return U.S. Equity... | U.S. Multifactor Equity  | FLAT       |
| 32 | VONV   | 0.0154 | Vanguard Russell 1000 Value ETF            | U.S. Large Cap Value     | LONG       |
| 38 | IUSV   | 0.0154 | iShares Core S&P U.S. Value ETF            | U.S. Large Cap Value     | FLAT       |
| 39 | SCHV   | 0.0154 | Schwab U.S. Large-Cap Value ETF            | U.S. Large Cap Value     | FLAT       |
| 42 | VTV    | 0.0154 | Vanguard Value ETF                         | U.S. Large Cap Value     | LONG       |
| 2  | RSP    | 0.0153 | Invesco S&P 500 Equal Weight ETF           | U.S. Equal Weight Equity | LONG       |
| 9  | PRF    | 0.0153 | Invesco RAFI US 1000 ETF                   | U.S. Fundamental Equity  | LONG       |
| 14 | CFA    | 0.0153 | VictoryShares US 500 Volatility Wtd ETF    | U.S. Multifactor Equity  | FLAT       |
| 16 | NULV   | 0.0153 | Nuveen ESG Large-Cap Value ETF             | U.S. Large Cap Value     | FLAT       |
| 20 | FNDX   | 0.0153 | Schwab Fundamental U.S. Large Company E... | U.S. Fundamental Equity  | LONG       |
| 26 | IVE    | 0.0153 | iShares S&P 500 Value ETF                  | U.S. Large Cap Value     | FLAT       |
| 30 | VOOV   | 0.0153 | Vanguard S&P 500 Value ETF                 | U.S. Large Cap Value     | FLAT       |
| 35 | VYM    | 0.0153 | Vanguard High Dividend Yield ETF           | U.S. Dividend Equity     | FLAT       |
| 36 | ILCV   | 0.0153 | iShares Morningstar Large-Cap Value ETF    | U.S. Large Cap Value     | FLAT       |
| 40 | SPYV   | 0.0153 | State Street SPDR Portfolio S&P 500 Val... | U.S. Large Cap Value     | FLAT       |
| 54 | VOE    | 0.0153 | Vanguard Mid-Cap Value ETF                 | U.S. Mid Cap Value       | FLAT       |
| 59 | DTD    | 0.0153 | WisdomTree U.S. Total Dividend Fund        | U.S. Dividend Equity     | LONG       |
| 64 | DGRO   | 0.0153 | iShares Core Dividend Growth ETF           | U.S. Dividend Equity     | LONG       |
| 22 | IWX    | 0.0152 | iShares Russell Top 200 Value ETF          | Value Equity             | LONG       |
| 33 | MGV    | 0.0152 | Vanguard Mega Cap Value ETF                | U.S. Mega Cap Value      | LONG       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 58 | DLN  | 0.0152 | WisdomTree U.S. LargeCap Dividend Fund     | U.S. Dividend Equity        | LONG |
| 10 | RWL  | 0.0151 | Invesco S&P 500 Revenue ETF                | U.S. Revenue Weighted Eq... | LONG |
| 12 | AIVL | 0.0151 | WisdomTree U.S. AI Enhanced Value Fund     | U.S. Value Factor Equity    | LONG |
| 23 | ONEV | 0.0151 | State Street SPDR Russell 1000 Low Vola... | U.S. Equity                 | FLAT |
| 43 | FTA  | 0.0151 | First Trust Large Cap Value AlphaDEX Fu... | Value Equity                | LONG |
| 51 | IWS  | 0.0151 | iShares Russell Mid-Cap Value ETF          | U.S. Mid Cap Value          | FLAT |
| 56 | PFM  | 0.0151 | Invesco Dividend Achievers ETF             | U.S. Dividend Equity        | FLAT |
| 67 | JPME | 0.0151 | JPMorgan Diversified Return U.S. Mid Ca... | U.S. Equity                 | FLAT |
| 21 | EQAL | 0.0150 | Invesco Russell 1000 Equal Weight ETF      | U.S. Equal Weight Equity    | FLAT |
| 52 | ONEY | 0.0150 | State Street SPDR Russell 1000 Yield Fo... | U.S. Dividend Equity        | FLAT |
| 5  | FEX  | 0.0149 | First Trust Large Cap Core AlphaDEX Fund   | Large Cap Blend Equities    | LONG |
| 18 | FDVV | 0.0149 | Fidelity High Dividend ETF                 | U.S. Dividend Equity        | FLAT |
| 34 | SCHD | 0.0149 | Schwab US Dividend Equity ETF              | U.S. Dividend Equity        | LONG |
| 65 | VIG  | 0.0149 | Vanguard Div Appreciation ETF              | U.S. Dividend Equity        | LONG |
| 17 | SDY  | 0.0148 | State Street SPDR S&P Dividend ETF         | U.S. Dividend Equity        | LONG |
| 28 | SIZE | 0.0148 | iShares MSCI USA Size Factor ETF           | U.S. Size Factor Equity     | LONG |
| 37 | IMCB | 0.0148 | iShares Morningstar Mid-Cap ETF            | U.S. Mid Cap Equity         | LONG |
| 4  | PKW  | 0.0147 | Invesco BuyBack Achievers ETF              | U.S. Buyback Equity         | LONG |
| 6  | PWV  | 0.0147 | Invesco Large Cap Value ETF                | Value Equity                | LONG |
| 25 | IWR  | 0.0147 | iShares Russell Mid-Cap ETF                | U.S. Mid Cap Equity         | LONG |
| 41 | VO   | 0.0147 | Vanguard Mid-Cap ETF                       | U.S. Mid Cap Equity         | LONG |
| 46 | SDOG | 0.0147 | ALPS Sector Dividend Dogs ETF              | U.S. Dividend Equity        | FLAT |
| 47 | DVY  | 0.0147 | iShares Select Dividend ETF                | U.S. Dividend Equity        | FLAT |
| 68 | JVAL | 0.0147 | JPMorgan U.S. Value Factor ETF             | U.S. Value Factor Equity    | LONG |
| 3  | FVD  | 0.0146 | First Trust VL Dividend                    | U.S. Dividend Equity        | LONG |
| 61 | DIA  | 0.0146 | State Street SPDR Dow Jones Industrial ... | U.S. Large Cap Equity       | LONG |
| 7  | RDVY | 0.0145 | First Trust Rising Dividend Achievers E... | U.S. Dividend Equity        | FLAT |
| 53 | SPYD | 0.0144 | State Street SPDR Portfolio S&P 500 Hig... | U.S. Dividend Equity        | FLAT |
| 57 | OUSL | 0.0144 | ALPS O'Shares U.S. Quality Dividend ETF    | Dividend Equity             | LONG |
| 63 | LGLV | 0.0144 | State Street SPDR US Large Cap Low Vola... | U.S. Low Volatility Equi... | LONG |
| 15 | CFO  | 0.0143 | VictoryShares US 500 Enhanced Volatilit... | U.S. Multifactor Equity     | FLAT |
| 49 | RPV  | 0.0143 | Invesco S&P 500 Pure Value ETF             | Value Equity                | FLAT |
| 55 | FTCS | 0.0143 | First Trust Capital Strength ETF           | U.S. Quality / Factor Eq... | LONG |
| 11 | DHS  | 0.0142 | WisdomTree U.S. High Dividend Fund         | U.S. Dividend Equity        | FLAT |
| 62 | USMV | 0.0141 | iShares MSCI USA Min Vol Factor ETF        | U.S. Low Volatility Equi... | LONG |
| 13 | CDC  | 0.0140 | VictoryShares US EQ Income Enhanced Vol... | U.S. Defensive Equity       | FLAT |
| 29 | VLUE | 0.0140 | iShares MSCI USA Value Factor ETF          | U.S. Value Factor Equity    | FLAT |
| 48 | RDIV | 0.0140 | Invesco S&P Ultra Dividend Revenue ETF     | U.S. Dividend Equity        | FLAT |
| 50 | SPHD | 0.0138 | Invesco S&P 500 High Dividend Low Volat... | Dividend Equity             | LONG |
| 8  | FDL  | 0.0137 | First Trust Morningstar ETF                | U.S. Dividend Equity        | FLAT |
| 44 | PEY  | 0.0137 | Invesco High Yield Equity Dividend Achi... | U.S. Dividend Equity        | FLAT |
| 1  | ACWV | 0.0135 | iShares MSCI Global Min Vol Factor ETF     | Global Equity               | LONG |
| 45 | DIV  | 0.0135 | Global X Super Dividend ETF                | U.S. Dividend Equity        | LONG |
| 19 | LVHD | 0.0134 | Franklin U.S. Low Volatility High Divid... | U.S. Low Volatility Equi... | LONG |
| 31 | HDV  | 0.0134 | iShares Core High Dividend ETF             | U.S. Dividend Equity        | LONG |

|    |      |        |                                     |                             |      |
|----|------|--------|-------------------------------------|-----------------------------|------|
| 60 | SPLV | 0.0129 | Invesco S&P 500 Low Volatility ETF  | U.S. Low Volatility Equi... | LONG |
| 66 | IYZ  | 0.0122 | iShares U.S. Telecommunications ETF | Communication Services      | FLAT |

### Cluster 3 – Developed ex-US Equity

**Members:** 61   **Var explained:** 0.894   **Signal:** LONG since 2026-07-06 (6 days, -1.3%)

#### Signal logic:

- Latest buy\_cross: 2026-07-06 – filter **PASS** (252d at date = +21.5%)
- Latest sell\_cross: 2026-06-25
- Current 252d return: +20.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-06 > last sell at 2026-06-25)



#### Category breakdown

| category               | n_members | abs_weight |
|------------------------|-----------|------------|
| International Equity   | 19        | 0.3158     |
| Developed ex-US Equity | 9         | 0.1528     |
| Global ex-US Equity    | 5         | 0.0848     |
| Europe Equity          | 4         | 0.0675     |

|                               |   |        |
|-------------------------------|---|--------|
| International Dividend Equity | 4 | 0.0659 |
| Alternatives                  | 4 | 0.0615 |
| Eurozone Equity               | 3 | 0.0490 |
| Commodity                     | 1 | 0.0172 |
| Global ex-US Small Cap Equity | 1 | 0.0166 |
| Germany Equity                | 1 | 0.0162 |
| France Equity                 | 1 | 0.0162 |
| Italy Equity                  | 1 | 0.0159 |
| Europe Financials             | 1 | 0.0158 |
| United Kingdom Equity         | 1 | 0.0158 |
| Netherlands Equity            | 1 | 0.0156 |
| Sweden Equity                 | 1 | 0.0156 |
| Spain Equity                  | 1 | 0.0151 |
| Austria Equity                | 1 | 0.0148 |
| Belgium Equity                | 1 | 0.0145 |
| Poland Equity                 | 1 | 0.0135 |

**Replication basket** – VEA (cor=0.99) basket lead at lag 1: -0.019

| ticker | rep_wt | name                                | category               | etf_signal |
|--------|--------|-------------------------------------|------------------------|------------|
| VEA    | 0.961  | Vanguard FTSE Developed Markets ETF | Developed ex-US Equity | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 2  | EFA    | 0.0173 | iShares MSCI EAFE ETF                      | Developed ex-US Equity      | LONG       |
| 51 | IEFA   | 0.0173 | iShares Core MSCI EAFE ETF                 | Developed ex-US Equity      | LONG       |
| 3  | VEA    | 0.0172 | Vanguard FTSE Developed Markets ETF        | Developed ex-US Equity      | FLAT       |
| 40 | GSIE   | 0.0172 | Goldman Sachs ActiveBeta International ... | Commodity                   | LONG       |
| 41 | ESGD   | 0.0172 | iShares ESG Aware MSCI EAFE ETF            | International Equity        | LONG       |
| 52 | SCHF   | 0.0172 | Schwab International Equity ETF            | Developed ex-US Equity      | FLAT       |
| 53 | SPDW   | 0.0172 | State Street SPDR Portfolio Developed W... | Developed ex-US Equity      | FLAT       |
| 25 | TLTD   | 0.0171 | FlexShares Morningstar Developed Market... | International Equity        | FLAT       |
| 6  | IEV    | 0.0170 | iShares Europe ETF                         | Europe Equity               | LONG       |
| 16 | DWM    | 0.0170 | WisdomTree International Equity Fund       | International Dividend E... | FLAT       |
| 19 | PXF    | 0.0170 | Invesco RAFI Developed Markets ex-U.S. ... | International Equity        | FLAT       |
| 29 | ACWX   | 0.0170 | iShares MSCI ACWI ex U.S. ETF              | Global ex-US Equity         | FLAT       |
| 32 | INTF   | 0.0170 | iShares International Equity Factor ETF    | International Equity        | LONG       |
| 46 | IEUR   | 0.0170 | iShares Core MSCI Europe ETF               | Europe Equity               | LONG       |
| 47 | IXUS   | 0.0170 | iShares Core MSCI Total International S... | Global ex-US Equity         | FLAT       |
| 48 | VEU    | 0.0170 | Vanguard FTSE All World Ex US ETF          | Global ex-US Equity         | FLAT       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 49 | VGK  | 0.0170 | Vanguard FTSEEuropean ETF                  | Europe Equity               | LONG |
| 50 | VXUS | 0.0170 | Vanguard Total International Stock ETF     | Global ex-US Equity         | FLAT |
| 30 | CWI  | 0.0169 | State Street SPDR MSCI ACWI ex-US ETF      | Global ex-US Equity         | FLAT |
| 33 | IQLT | 0.0169 | iShares MSCI Intl Quality Factor ETF       | International Equity        | LONG |
| 35 | QEFA | 0.0169 | State Street SPDR MSCI EAFE StrategicFa... | International Equity        | LONG |
| 39 | FNDF | 0.0169 | Schwab Fundamental International Equity... | International Equity        | FLAT |
| 14 | EZU  | 0.0168 | iShares MSCI Eurozone ETF                  | Eurozone Equity             | FLAT |
| 23 | EFV  | 0.0168 | iShares MSCI EAFE Value ETF                | Developed ex-US Equity      | LONG |
| 26 | JPIN | 0.0168 | JPMorgan Diversified Return Internation... | International Equity        | FLAT |
| 38 | VYMI | 0.0168 | Vanguard International High Dividend Yi... | International Dividend E... | LONG |
| 18 | IQDF | 0.0167 | FlexShares International Quality Divide... | International Equity        | FLAT |
| 22 | SCZ  | 0.0167 | iShares MSCI EAFE Small-Cap ETF            | Developed ex-US Equity      | FLAT |
| 24 | FNDC | 0.0167 | Schwab Fundamental International Small ... | International Equity        | FLAT |
| 37 | RODM | 0.0167 | Hartford Multifactor Developed Markets ... | International Equity        | FLAT |
| 44 | SCHC | 0.0167 | Schwab International Small-Cap Equity E... | Developed ex-US Equity      | FLAT |
| 56 | HFXI | 0.0167 | NYLI FTSE International Equity Currency... | International Equity        | FLAT |
| 20 | EFG  | 0.0166 | iShares MSCI EAFE Growth ETF               | Developed ex-US Equity      | FLAT |
| 34 | IVLU | 0.0166 | iShares MSCI Intl Value Factor ETF         | International Equity        | LONG |
| 36 | FEZ  | 0.0166 | State Street SPDR EURO STOXX 50 ETF        | Eurozone Equity             | FLAT |
| 45 | VSS  | 0.0166 | Vanguard FTSE All-Wld ex-US SmCp Idx ETF   | Global ex-US Small Cap E... | FLAT |
| 4  | FEP  | 0.0165 | First Trust Europe AlphaDEX Fund           | Europe Equity               | FLAT |
| 7  | DLS  | 0.0165 | WisdomTree International SmallCap Fund     | International Equity        | FLAT |
| 31 | IMTM | 0.0164 | iShares MSCI Intl Momentum Factor ETF      | International Equity        | FLAT |
| 43 | VIGI | 0.0163 | Vanguard International Dividend Appreci... | International Dividend E... | LONG |
| 1  | EWG  | 0.0162 | iShares MSCI Germany Index Fund            | Germany Equity              | FLAT |
| 12 | EWQ  | 0.0162 | iShares MSCI France Index Fund             | France Equity               | FLAT |
| 21 | GWX  | 0.0162 | State Street SPDR S&P International Sma... | International Equity        | FLAT |
| 5  | VIDI | 0.0160 | Vident International Equity Strategy ETF   | International Equity        | FLAT |
| 59 | IPKW | 0.0160 | Invesco International BuyBack Achievers... | International Equity        | LONG |
| 61 | EWI  | 0.0159 | iShares MSCI Italy ETF                     | Italy Equity                | FLAT |
| 13 | EWU  | 0.0158 | iShares MSCI United Kingdom ETF            | United Kingdom Equity       | LONG |
| 15 | IDV  | 0.0158 | iShares International Select Dividend E... | International Dividend E... | FLAT |
| 17 | EUFN | 0.0158 | iShares MSCI Europe Financials ETF         | Europe Financials           | LONG |
| 27 | DBEF | 0.0157 | Xtrackers MSCI EAFE Hedged Equity ETF      | Alternatives                | FLAT |
| 42 | HDEF | 0.0157 | Xtrackers MSCI EAFE High Dividend Yield... | International Equity        | LONG |
| 10 | HEZU | 0.0156 | iShares Currency Hedged MSCI Eurozone E... | Eurozone Equity             | FLAT |
| 28 | HEFA | 0.0156 | iShares Currency Hedged MSCI EAFE ETF      | Alternatives                | FLAT |
| 55 | EWD  | 0.0156 | iShares MSCI Sweden ETF                    | Sweden Equity               | FLAT |
| 58 | EWN  | 0.0156 | iShares MSCI Netherlands Index Fund        | Netherlands Equity          | FLAT |
| 8  | HEDJ | 0.0151 | WisdomTree Europe Hedged Equity Fund       | Alternatives                | FLAT |
| 9  | IHDG | 0.0151 | WisdomTree International Hedged Quality... | Alternatives                | FLAT |
| 11 | EWI  | 0.0151 | iShares MSCI Spain ETF                     | Spain Equity                | FLAT |
| 57 | EWO  | 0.0148 | iShares MSCI Austria ETF                   | Austria Equity              | LONG |
| 60 | EWK  | 0.0145 | iShares MSCI Belgium ETF                   | Belgium Equity              | FLAT |
| 54 | EPOL | 0.0135 | iShares MSCI Poland ETF                    | Poland Equity               | LONG |

**Cluster 4 – Aggregate Fixed Income****Members:** 58   **Var explained:** 0.832   **Signal:** FLAT**Signal logic:**

- Latest buy\_cross: 2026-06-17 – filter **PASS** (252d at date = +4.5%)
- Latest sell\_cross: 2026-07-07
- Current 252d return: +3.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)

**Category breakdown**

| category                        | n_members | abs_weight |
|---------------------------------|-----------|------------|
| Fixed Income                    | 19        | 0.3284     |
| Investment Grade Corporate Bond | 13        | 0.2302     |
| Aggregate Fixed Income          | 6         | 0.1120     |
| Mortgage-Backed Fixed Income    | 6         | 0.0982     |
| Active Fixed Income             | 5         | 0.0891     |
| Cash / Ultra-Short Fixed Income | 3         | 0.0471     |
| International Fixed Income      | 2         | 0.0313     |
| Target Maturity Bond            | 2         | 0.0306     |
| Agency Fixed Income             | 1         | 0.0167     |
| Taxable Municipal Bond          | 1         | 0.0165     |

**Replication basket** – BND (cor=0.99) basket lead at lag 1: 0.006

| ticker | rep_wt | name                           | category               | etf_signal |
|--------|--------|--------------------------------|------------------------|------------|
| BND    | 1.007  | Vanguard Total Bond Market ETF | Aggregate Fixed Income | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                        | category                    | etf_signal |
|----|--------|--------|---------------------------------------------|-----------------------------|------------|
| 21 | BIV    | 0.0188 | Vanguard Intermediate-Term Bond ETF         | Fixed Income                | FLAT       |
| 26 | AGG    | 0.0188 | iShares Core U.S. Aggregate Bond ETF        | Aggregate Fixed Income      | FLAT       |
| 28 | BND    | 0.0188 | Vanguard Total Bond Market ETF              | Aggregate Fixed Income      | FLAT       |
| 30 | SCHZ   | 0.0188 | Schwab US Aggregate Bond ETF                | Aggregate Fixed Income      | FLAT       |
| 20 | IUSB   | 0.0187 | iShares Core Universal USD Bond ETF         | Aggregate Fixed Income      | FLAT       |
| 31 | SPAB   | 0.0187 | State Street SPDR Portfolio Aggregate B...  | Aggregate Fixed Income      | FLAT       |
| 2  | IEF    | 0.0185 | iShares 7-10 Year Treasury Bond ETF         | Fixed Income                | FLAT       |
| 17 | IGIB   | 0.0184 | iShares 5-10 Year Investment Grade Corp...  | Investment Grade Corpora... | FLAT       |
| 24 | VCIT   | 0.0184 | Vanguard Intermediate-Term Corporate Bo...  | Investment Grade Corpora... | FLAT       |
| 7  | FBND   | 0.0183 | Fidelity Total Bond ETF                     | Active Fixed Income         | FLAT       |
| 43 | SPTI   | 0.0183 | State Street SPDR Portfolio Intermediat...  | Fixed Income                | FLAT       |
| 46 | SCHR   | 0.0183 | Schwab Intermediate-Term U.S. Treasury ...  | Fixed Income                | FLAT       |
| 47 | VGIT   | 0.0183 | Vanguard Intermediate-Term Treasury ETF     | Fixed Income                | FLAT       |
| 13 | AGGY   | 0.0182 | WisdomTree Yield Enhanced U.S. Aggregat...  | Aggregate Fixed Income      | FLAT       |
| 15 | SPIB   | 0.0182 | State Street SPDR Portfolio Intermediat...  | Investment Grade Corpora... | FLAT       |
| 6  | GTO    | 0.0181 | Invesco Total Return Bond ETF               | Active Fixed Income         | FLAT       |
| 11 | GVI    | 0.0181 | iShares Intermediate Government/Credit ...  | Fixed Income                | FLAT       |
| 40 | IEI    | 0.0181 | iShares 3-7 Year Treasury Bond ETF          | Fixed Income                | FLAT       |
| 50 | MBB    | 0.0181 | iShares MBS ETF                             | Mortgage-Backed Fixed In... | FLAT       |
| 12 | QLTA   | 0.0180 | iShares Aaa A Rated Corporate Bond ETF      | Investment Grade Corpora... | FLAT       |
| 27 | USIG   | 0.0180 | iShares Broad USD Investment Grade Corp...  | Investment Grade Corpora... | FLAT       |
| 51 | VMBS   | 0.0180 | Vanguard Mortgage-Backed Securities ETF     | Mortgage-Backed Fixed In... | FLAT       |
| 52 | SPMB   | 0.0180 | State Street SPDR Portfolio Mortgage Ba...  | Mortgage-Backed Fixed In... | FLAT       |
| 3  | LQD    | 0.0179 | iShares iBoxx \$ Investment Grade Corpor... | Investment Grade Corpora... | FLAT       |
| 5  | BOND   | 0.0179 | PIMCO Active Bond Exchange-Traded Fund ...  | Active Fixed Income         | FLAT       |
| 19 | ISTB   | 0.0178 | iShares Core 1-5 Year USD Bond ETF          | Fixed Income                | FLAT       |
| 8  | FLCO   | 0.0177 | Franklin Investment Grade Corporate ETF     | Active Fixed Income         | FLAT       |
| 10 | CORP   | 0.0177 | PIMCO Investment Grade Corporate Bond I...  | Investment Grade Corpora... | FLAT       |
| 22 | BLV    | 0.0176 | Vanguard Long-Term Bond ETF                 | Fixed Income                | FLAT       |
| 41 | TLH    | 0.0176 | iShares 10-20 Year Treasury Bond ETF        | Fixed Income                | FLAT       |
| 45 | GOVT   | 0.0176 | iShares U.S. Treasury Bond ETF              | Fixed Income                | FLAT       |
| 35 | IGSB   | 0.0175 | iShares 1-5 Year Investment Grade Corpo...  | Investment Grade Corpora... | FLAT       |
| 37 | VCSH   | 0.0175 | Vanguard Short-Term Corporate Bond ETF      | Investment Grade Corpora... | FLAT       |
| 23 | BSV    | 0.0174 | Vanguard Short-Term Bond ETF                | Fixed Income                | FLAT       |

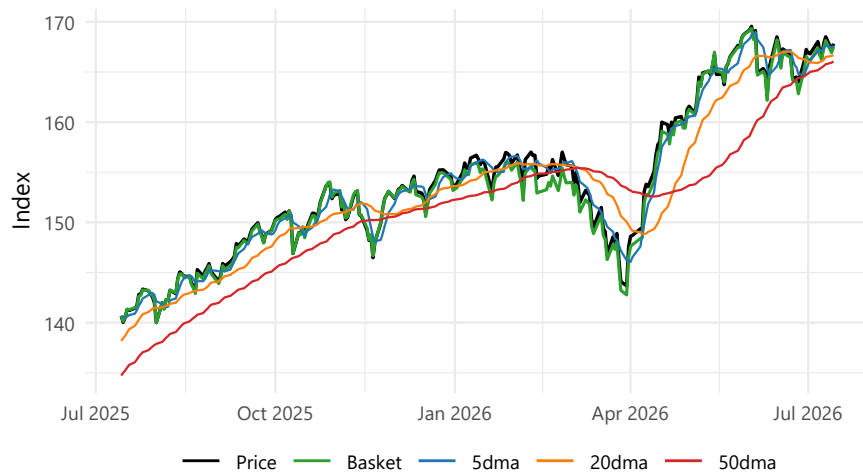
|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 49 | GNMA | 0.0174 | iShares GNMA Bond ETF                      | Mortgage-Backed Fixed In... | FLAT |
| 18 | IGLB | 0.0172 | iShares 10 Year Investment Grade Corpor... | Investment Grade Corpora... | FLAT |
| 16 | SPLB | 0.0171 | State Street SPDR Portfolio Long Term C... | Investment Grade Corpora... | FLAT |
| 25 | VCLT | 0.0171 | Vanguard Long-Term Corporate Bond ETF      | Investment Grade Corpora... | FLAT |
| 36 | SLQD | 0.0171 | iShares 0-5 Year Investment Grade Corpo... | Investment Grade Corpora... | FLAT |
| 44 | SPTL | 0.0171 | State Street SPDR Portfolio Long Term T... | Fixed Income                | FLAT |
| 48 | VGLT | 0.0171 | Vanguard Long-Term Treasury ETF            | Fixed Income                | FLAT |
| 29 | TOTL | 0.0170 | State Street DoubleLine Total Return Ta... | Active Fixed Income         | FLAT |
| 1  | TLT  | 0.0169 | iShares 20+ Year Treasury Bond ETF         | Fixed Income                | FLAT |
| 39 | AGZ  | 0.0167 | iShares Agency Bond ETF                    | Agency Fixed Income         | FLAT |
| 9  | BAB  | 0.0165 | Invesco Taxable Municipal Bond ETF         | Taxable Municipal Bond      | FLAT |
| 42 | EDV  | 0.0163 | Vanguard Extended Duration Treasury ETF    | Fixed Income                | FLAT |
| 34 | SPSB | 0.0162 | State Street SPDR Portfolio Short Term ... | Fixed Income                | FLAT |
| 53 | SHY  | 0.0162 | iShares 1-3 Year Treasury Bond ETF         | Fixed Income                | FLAT |
| 56 | VGSH | 0.0159 | Vanguard Short-Term Treasury ETF           | Cash / Ultra-Short Fixed... | FLAT |
| 57 | BNDX | 0.0158 | Vanguard Total International Bond ETF      | International Fixed Inco... | FLAT |
| 33 | BSCQ | 0.0156 | Invesco BulletShares 2026 Corporate Bon... | Target Maturity Bond        | LONG |
| 55 | SCHO | 0.0156 | Schwab Short-Term U.S. Treasury ETF        | Cash / Ultra-Short Fixed... | FLAT |
| 58 | IAGG | 0.0156 | iShares International Aggregate Bond Fu... | International Fixed Inco... | FLAT |
| 54 | SPTS | 0.0155 | State Street SPDR Portfolio Short Term ... | Cash / Ultra-Short Fixed... | FLAT |
| 14 | IBDR | 0.0151 | iShares iBonds Dec 2026 Term Corporate ... | Target Maturity Bond        | FLAT |
| 38 | CMBS | 0.0136 | iShares CMBS Bond ETF                      | Mortgage-Backed Fixed In... | FLAT |
| 4  | LMBS | 0.0133 | First Trust Low Duration Opportunities ... | Mortgage-Backed Fixed In... | FLAT |
| 32 | LDUR | 0.0121 | PIMCO Enhanced Low Duration Active Exch... | Fixed Income                | FLAT |

## Cluster 5 – U.S. Large Cap Equity

**Members:** 47   **Var explained:** 0.948   **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-04-13 – filter **PASS** (252d at date = +25.6%)
- Latest sell\_cross: 2026-06-24
- Current 252d return: +19.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-24 more recent than last effective buy 2026-04-13)



### Category breakdown

| category                     | n_members | abs_weight |
|------------------------------|-----------|------------|
| U.S. Large Cap Equity        | 9         | 0.1959     |
| U.S. Total Market Equity     | 7         | 0.1525     |
| Global Equity                | 7         | 0.1495     |
| Asset Allocation             | 4         | 0.0792     |
| U.S. Mega Cap Equity         | 3         | 0.0640     |
| U.S. Multifactor Equity      | 3         | 0.0634     |
| U.S. Quality / Factor Equity | 3         | 0.0634     |
| U.S. ESG Equity              | 2         | 0.0434     |
| U.S. Equity                  | 2         | 0.0434     |
| Commodity                    | 1         | 0.0217     |
| Large Cap Value Equities     | 1         | 0.0216     |
| Large Cap Blend Equities     | 1         | 0.0215     |
| Dividend Equity              | 1         | 0.0213     |
| U.S. Dividend Equity         | 1         | 0.0209     |
| U.S. Moat Equity             | 1         | 0.0198     |
| Alternatives                 | 1         | 0.0186     |

**Replication basket** – VOO (cor=1.00) basket lead at lag 1: -0.010

| ticker | rep_wt | name | category | etf_signal |
|--------|--------|------|----------|------------|
|--------|--------|------|----------|------------|

V00 0.936 Vanguard S&P 500 ETF U.S. Large Cap Equity LONG

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 1  | SPY    | 0.0218 | State Street SPDR S&P 500 ETF Trust        | U.S. Large Cap Equity       | LONG       |
| 21 | IWV    | 0.0218 | iShares Russell 3000 Fund                  | U.S. Total Market Equity    | FLAT       |
| 22 | IYY    | 0.0218 | iShares Dow Jones U.S. ETF                 | U.S. Total Market Equity    | FLAT       |
| 27 | ESGU   | 0.0218 | iShares ESG Aware MSCI USA ETF             | U.S. ESG Equity             | FLAT       |
| 28 | IWB    | 0.0218 | iShares Russell 1000 ETF                   | U.S. Large Cap Equity       | FLAT       |
| 36 | VONE   | 0.0218 | Vanguard Russell 1000 ETF                  | U.S. Large Cap Equity       | FLAT       |
| 38 | VV     | 0.0218 | Vanguard Large-Cap ETF                     | U.S. Large Cap Equity       | FLAT       |
| 39 | ILCB   | 0.0218 | iShares Morningstar Large-Cap ETF          | U.S. Large Cap Equity       | FLAT       |
| 40 | ITOT   | 0.0218 | iShares Core S&P Total U.S. Stock Marke... | U.S. Total Market Equity    | FLAT       |
| 41 | IVV    | 0.0218 | iShares Core S&P 500 ETF                   | U.S. Large Cap Equity       | LONG       |
| 42 | SCHB   | 0.0218 | Schwab U.S. Broad Market ETF               | U.S. Total Market Equity    | FLAT       |
| 43 | SCHX   | 0.0218 | Schwab U.S. Large-Cap ETF                  | U.S. Large Cap Equity       | FLAT       |
| 44 | SPTM   | 0.0218 | State Street SPDR Portfolio S&P 1500 Co... | U.S. Total Market Equity    | FLAT       |
| 45 | V00    | 0.0218 | Vanguard S&P 500 ETF                       | U.S. Large Cap Equity       | LONG       |
| 46 | VTI    | 0.0218 | Vanguard Total Stock Market ETF            | U.S. Total Market Equity    | FLAT       |
| 8  | URTH   | 0.0217 | iShares MSCI World ETF                     | Global Equity               | LONG       |
| 20 | SUSA   | 0.0217 | iShares ESG Optimized MSCI USA ETF         | U.S. ESG Equity             | FLAT       |
| 25 | SPYX   | 0.0217 | State Street SPDR S&P 500 Fossil Fuel R... | U.S. Equity                 | LONG       |
| 33 | VTHR   | 0.0217 | Vanguard Russell 3000 ETF                  | U.S. Total Market Equity    | FLAT       |
| 34 | GSLC   | 0.0217 | Goldman Sachs ActiveBeta U.S. Large Cap... | Commodity                   | LONG       |
| 17 | JHML   | 0.0216 | John Hancock Multifactor Large Cap ETF     | U.S. Large Cap Equity       | FLAT       |
| 23 | LRGF   | 0.0216 | iShares U.S. Equity Factor ETF             | U.S. Multifactor Equity     | FLAT       |
| 29 | IWL    | 0.0216 | iShares Russell Top 200 ETF                | U.S. Equity                 | LONG       |
| 35 | EPS    | 0.0216 | WisdomTree U.S. LargeCap Fund              | Large Cap Value Equities    | FLAT       |
| 37 | MGC    | 0.0216 | Vanguard Mega Cap ETF                      | U.S. Mega Cap Equity        | LONG       |
| 3  | ACWI   | 0.0215 | iShares MSCI ACWI ETF                      | Global Equity               | FLAT       |
| 11 | VT     | 0.0215 | Vanguard Total World Stock Index ETF       | Global Equity               | FLAT       |
| 13 | TILT   | 0.0215 | FlexShares Mornigstar US Market Factors... | U.S. Multifactor Equity     | FLAT       |
| 19 | DSI    | 0.0215 | iShares ESG MSCI KLD 400 ETF               | Large Cap Blend Equities    | FLAT       |
| 9  | CRBN   | 0.0214 | iShares Low Carbon Optimized MSCI ACWI ... | Global Equity               | LONG       |
| 10 | SPGM   | 0.0214 | State Street SPDR Portfolio MSCI Global... | Global Equity               | FLAT       |
| 24 | OEF    | 0.0214 | iShares S&P 100 Fund                       | U.S. Mega Cap Equity        | LONG       |
| 30 | QUAL   | 0.0214 | iShares MSCI USA Quality Factor ETF        | U.S. Quality / Factor Eq... | FLAT       |
| 4  | AOA    | 0.0213 | iShares Core 80/20 Aggressive Allocatio... | Asset Allocation            | FLAT       |
| 15 | QDF    | 0.0213 | FlexShares Quality Dividend Index Fund     | Dividend Equity             | FLAT       |
| 14 | IOO    | 0.0211 | iShares Global 100 ETF                     | Global Equity               | FLAT       |
| 31 | QUS    | 0.0211 | State Street SPDR MSCI USA StrategicFac... | U.S. Quality / Factor Eq... | FLAT       |
| 16 | CATH   | 0.0210 | Global X S&P 500 Catholic Values ETF       | Global Equity               | LONG       |

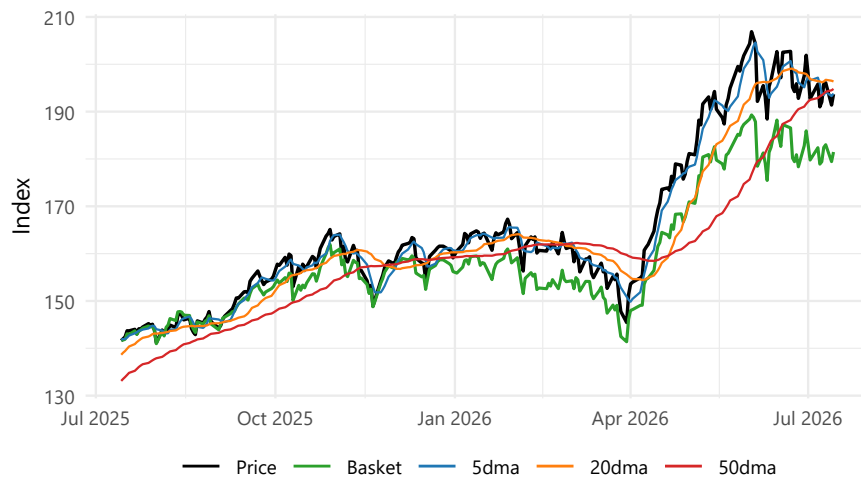
|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 26 | XLG  | 0.0210 | Invesco S&P 500 Top 50 ETF                 | U.S. Mega Cap Equity        | FLAT |
| 18 | DGRW | 0.0209 | WisdomTree U.S. Quality Dividend Growth... | U.S. Dividend Equity        | LONG |
| 32 | SPHQ | 0.0208 | Invesco S&P 500 Quality ETF                | U.S. Quality / Factor Eq... | FLAT |
| 7  | AOR  | 0.0207 | iShares Core 60/40 Balanced Allocation ... | Asset Allocation            | FLAT |
| 47 | OMFL | 0.0203 | Invesco Russell 1000 Dynamic Multifacto... | U.S. Multifactor Equity     | FLAT |
| 12 | MOAT | 0.0198 | VanEck Morningstar Wide Moat ETF           | U.S. Moat Equity            | FLAT |
| 6  | AOM  | 0.0194 | iShares Core 40/60 Moderate Allocation ... | Asset Allocation            | FLAT |
| 2  | QAI  | 0.0186 | NYLI Hedge Multi-Strategy Tracker ETF      | Alternatives                | FLAT |
| 5  | AOK  | 0.0178 | iShares Core 30/70 Conservative Allocat... | Asset Allocation            | FLAT |

### Cluster 6 – U.S. Large Cap Growth

**Members:** 44   **Var explained:** 0.883   **Signal:** FLAT

#### Signal logic:

- Latest buy\_cross: 2026-04-10 – filter **PASS** (252d at date = +54.9%)
- Latest sell\_cross: 2026-07-08
- Current 252d return: +36.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-08 more recent than last effective buy 2026-04-10)



#### Category breakdown

| category                    | n_members | abs_weight |
|-----------------------------|-----------|------------|
| Technology                  | 23        | 0.5198     |
| U.S. Large Cap Growth       | 11        | 0.2593     |
| Growth Equity               | 2         | 0.0464     |
| U.S. Equity                 | 2         | 0.0463     |
| U.S. Mega Cap Growth        | 1         | 0.0234     |
| U.S. Equal Weight Equity    | 1         | 0.0230     |
| Options Income              | 1         | 0.0213     |
| U.S. Small Cap Technology   | 1         | 0.0213     |
| Trend-Following Equity      | 1         | 0.0209     |
| U.S. Trend-Following Equity | 1         | 0.0183     |

**Replication basket** – QQQ (cor=0.98) basket lead at lag 1: -0.027

| ticker | rep_wt | name                        | category              | etf_signal |
|--------|--------|-----------------------------|-----------------------|------------|
| QQQ    | 1.017  | Invesco QQQ Trust, Series 1 | U.S. Large Cap Growth | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category              | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------|------------|
| 21 | IGM    | 0.0239 | iShares Expanded Tech Sector ETF           | Technology            | FLAT       |
| 7  | QQQ    | 0.0238 | Invesco QQQ Trust, Series 1                | U.S. Large Cap Growth | FLAT       |
| 23 | IYW    | 0.0238 | iShares U.S. Technology ETF                | Technology            | FLAT       |
| 25 | ONEQ   | 0.0238 | Fidelity Nasdaq Composite Index ETF        | U.S. Equity           | LONG       |
| 29 | VGT    | 0.0238 | Vanguard Information Tech ETF              | Technology            | FLAT       |
| 31 | FTEC   | 0.0238 | Fidelity MSCI Information Technology In... | Technology            | FLAT       |
| 34 | ILCG   | 0.0237 | iShares Morningstar Large-Cap Growth ETF   | U.S. Large Cap Growth | LONG       |
| 1  | XLK    | 0.0236 | State Street Technology Select Sector S... | Technology            | FLAT       |
| 22 | IXN    | 0.0236 | iShares Global Tech ETF                    | Technology            | FLAT       |
| 35 | IUSG   | 0.0236 | iShares Core S&P U.S. Growth ETF           | U.S. Large Cap Growth | LONG       |
| 17 | XNTK   | 0.0235 | State Street SPDR NYSE Technology ETF      | Technology            | FLAT       |
| 24 | NULG   | 0.0235 | Nuveen ESG Large-Cap Growth ETF            | U.S. Large Cap Growth | FLAT       |
| 27 | IWF    | 0.0235 | iShares Russell 1000 Growth Fund           | U.S. Large Cap Growth | FLAT       |
| 28 | IVW    | 0.0235 | iShares S&P 500 Growth ETF                 | U.S. Large Cap Growth | LONG       |
| 30 | VOOG   | 0.0235 | Vanguard S&P 500 Growth ETF                | U.S. Large Cap Growth | LONG       |
| 32 | VONG   | 0.0235 | Vanguard Russell 1000 Growth ETF           | U.S. Large Cap Growth | FLAT       |
| 37 | SPYG   | 0.0235 | State Street SPDR Portfolio S&P 500 Gro... | U.S. Large Cap Growth | LONG       |
| 38 | VUG    | 0.0235 | Vanguard Growth ETF                        | U.S. Large Cap Growth | LONG       |
| 11 | QTEC   | 0.0234 | First Trust NASDAQ-100-Technology Secto... | Technology            | FLAT       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 33 | MGK  | 0.0234 | Vanguard Mega Cap Growth ETF               | U.S. Mega Cap Growth        | LONG |
| 36 | SCHG | 0.0234 | Schwab U.S. Large-Cap Growth ETF           | U.S. Large Cap Growth       | LONG |
| 16 | FXL  | 0.0233 | First Trust Technology AlphaDEX            | Technology                  | FLAT |
| 26 | IWY  | 0.0233 | iShares Russell Top 200 Growth ETF         | Growth Equity               | FLAT |
| 6  | AIQ  | 0.0232 | Global X Artificial Intelligence & Tech... | Technology                  | FLAT |
| 20 | PWB  | 0.0231 | Invesco Large Cap Growth ETF               | Growth Equity               | FLAT |
| 10 | QQEW | 0.0230 | First Trust Nasdaq-100 Select Equal Wei... | U.S. Equal Weight Equity    | LONG |
| 14 | XT   | 0.0228 | iShares Future Exponential Technologies... | Technology                  | FLAT |
| 13 | TDIV | 0.0226 | First Trust NASDAQ Technology Dividend ... | Technology                  | FLAT |
| 39 | SPHB | 0.0225 | Invesco S&P 500 High Beta ETF              | U.S. Equity                 | FLAT |
| 3  | SMH  | 0.0224 | VanEck Semiconductor ETF                   | Technology                  | FLAT |
| 2  | SOXX | 0.0222 | iShares PHLX SOX Semiconductor Sector I... | Technology                  | FLAT |
| 44 | QTUM | 0.0222 | Defiance Quantum ETF                       | Technology                  | LONG |
| 5  | ROBO | 0.0221 | ROBO Global Robotics and Automation Ind... | Technology                  | FLAT |
| 4  | BOTZ | 0.0220 | Global X Robotics & Artificial Intellig... | Technology                  | FLAT |
| 15 | XSD  | 0.0218 | State Street SPDR S&P Semiconductor ETF    | Technology                  | FLAT |
| 8  | NXTG | 0.0217 | First Trust Indxx NextG ETF                | Technology                  | FLAT |
| 12 | PSI  | 0.0217 | Invesco Semiconductors ETF                 | Technology                  | FLAT |
| 42 | FTXL | 0.0217 | First Trust Nasdaq Semiconductor ETF       | Technology                  | FLAT |
| 43 | PTF  | 0.0214 | Invesco Dorsey Wright Technology Moment... | Technology                  | LONG |
| 19 | QYLD | 0.0213 | Global X NASDAQ 100 Covered Call ETF       | Options Income              | FLAT |
| 41 | PSCT | 0.0213 | Invesco S&P SmallCap Information Techno... | U.S. Small Cap Technology   | FLAT |
| 18 | PTNQ | 0.0209 | Pacer Trendpilot 100 ETF                   | Trend-Following Equity      | FLAT |
| 9  | PRNT | 0.0194 | 3D Printing (The) ETF                      | Technology                  | FLAT |
| 40 | PTLC | 0.0183 | Pacer Trendpilot US Large Cap ETF          | U.S. Trend-Following Equ... | FLAT |

## Cluster 7 – U.S. Extended Market Equity

**Members:** 42   **Var explained:** 0.782   **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-04-13 – filter **PASS** (252d at date = +25.8%)
- Latest sell\_cross: 2026-06-10
- Current 252d return: +15.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-10 more recent than last effective buy 2026-04-13)



### Category breakdown

| category                       | n_members | abs_weight |
|--------------------------------|-----------|------------|
| Technology                     | 12        | 0.2792     |
| U.S. Mid Cap Growth            | 3         | 0.0786     |
| U.S. Small Cap Growth          | 3         | 0.0774     |
| U.S. Momentum Equity           | 3         | 0.0698     |
| Communication Services         | 3         | 0.0676     |
| Growth Equity                  | 2         | 0.0508     |
| IPO Equity                     | 2         | 0.0499     |
| Convertible Bond               | 2         | 0.0493     |
| Momentum Equity                | 2         | 0.0479     |
| Financials                     | 2         | 0.0478     |
| Consumer Discretionary         | 2         | 0.0454     |
| U.S. Extended Market Equity    | 1         | 0.0262     |
| U.S. Small Cap Momentum Equity | 1         | 0.0235     |
| U.S. Mid Cap Momentum Equity   | 1         | 0.0234     |
| Industrials                    | 1         | 0.0222     |
| International IPO Equity       | 1         | 0.0211     |
| Israel Equity                  | 1         | 0.0198     |

**Replication basket** – VXF (cor=0.97) basket lead at lag 1: 0.013

| ticker | rep_wt | name                         | category                 | etf_signal |
|--------|--------|------------------------------|--------------------------|------------|
| VXF    | 0.988  | Vanguard Extended Market ETF | U.S. Extended Market ... | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 25 | IWP    | 0.0263 | iShares Russell Midcap Growth ETF          | U.S. Mid Cap Growth         | LONG       |
| 30 | VOT    | 0.0262 | Vanguard Mid-Cap Growth ETF                | U.S. Mid Cap Growth         | LONG       |
| 35 | VBK    | 0.0262 | Vanguard Small-Cap Growth ETF              | U.S. Small Cap Growth       | FLAT       |
| 37 | VXF    | 0.0262 | Vanguard Extended Market ETF               | U.S. Extended Market Equ... | FLAT       |
| 31 | IMCG   | 0.0261 | iShares Morningstar Mid-Cap Growth ETF     | U.S. Mid Cap Growth         | LONG       |
| 36 | ISCG   | 0.0258 | iShares Morningstar Small-Cap Growth ETF   | Growth Equity               | FLAT       |
| 33 | IWO    | 0.0256 | iShares Russell 2000 Growth Fund           | U.S. Small Cap Growth       | FLAT       |
| 34 | VTWG   | 0.0256 | Vanguard Russell 2000 Growth ETF           | U.S. Small Cap Growth       | FLAT       |
| 21 | FPX    | 0.0254 | First Trust US Equity Opportunities ETF    | IPO Equity                  | FLAT       |
| 19 | FTC    | 0.0252 | First Trust Large Cap Growth AlphaDEX F... | Technology                  | FLAT       |
| 23 | RPG    | 0.0250 | Invesco S&P 500 Pure Growth ETF            | Growth Equity               | FLAT       |
| 22 | CWB    | 0.0249 | State Street SPDR Bloomberg Convertible... | Convertible Bond            | FLAT       |
| 20 | PDP    | 0.0247 | Invesco Dorsey Wright Momentum ETF         | Momentum Equity             | FLAT       |
| 13 | IPO    | 0.0245 | Renaissance IPO ETF                        | IPO Equity                  | FLAT       |
| 18 | FINX   | 0.0245 | Global X FinTech ETF                       | Financials                  | FLAT       |
| 39 | VFMO   | 0.0245 | Vanguard U.S. Momentum Factor ETF          | U.S. Momentum Equity        | FLAT       |
| 8  | ARKQ   | 0.0244 | ARK Autonomous Technology & Robotics ETF   | Technology                  | FLAT       |
| 15 | FDN    | 0.0244 | First Trust DJ Internet Index Fund         | Technology                  | LONG       |
| 26 | ICVT   | 0.0244 | iShares Convertible Bond ETF               | Convertible Bond            | FLAT       |
| 7  | ARKW   | 0.0241 | ARK Next Generation Internet ETF           | Technology                  | LONG       |
| 3  | SKYY   | 0.0240 | First Trust Cloud Computing ETF            | Technology                  | LONG       |
| 6  | ARKK   | 0.0240 | ARK Innovation ETF                         | Technology                  | LONG       |
| 14 | PNQI   | 0.0240 | Invesco Nasdaq Internet ETF                | Technology                  | FLAT       |
| 11 | IBUY   | 0.0238 | Amplify Online Retail ETF                  | Consumer Discretionary      | LONG       |
| 32 | DWAS   | 0.0235 | Invesco Dorsey Wright SmallCap Momentum... | U.S. Small Cap Momentum ... | FLAT       |
| 16 | IPAY   | 0.0234 | Amplify Digital Payments ETF               | Financials                  | FLAT       |
| 24 | XMMO   | 0.0234 | Invesco S&P MidCap Momentum ETF            | U.S. Mid Cap Momentum Eq... | FLAT       |
| 10 | FV     | 0.0232 | First Trust Dorsey Wright Focus 5 ETF      | Momentum Equity             | FLAT       |
| 2  | IGV    | 0.0230 | iShares Expanded Tech-Software Sector E... | Technology                  | FLAT       |
| 27 | MTUM   | 0.0230 | iShares MSCI USA Momentum Factor ETF       | U.S. Momentum Equity        | FLAT       |
| 28 | VOX    | 0.0229 | Vanguard Communication Services ETF        | Communication Services      | FLAT       |
| 29 | FCOM   | 0.0229 | Fidelity MSCI Communication Services In... | Communication Services      | FLAT       |
| 12 | CIBR   | 0.0225 | First Trust NASDAQ Cybersecurity ETF       | Technology                  | FLAT       |
| 5  | HACK   | 0.0223 | Amplify Cybersecurity ETF                  | Technology                  | FLAT       |
| 38 | QMOM   | 0.0223 | Alpha Architect U.S. Quantitative Momen... | U.S. Momentum Equity        | FLAT       |
| 42 | PRN    | 0.0222 | Invesco Dorsey Wright Industrials Momen... | Industrials                 | FLAT       |

|    |      |        |                                            |                          |      |
|----|------|--------|--------------------------------------------|--------------------------|------|
| 4  | BLOK | 0.0220 | Amplify Blockchain Technology ETF          | Technology               | FLAT |
| 1  | XLC  | 0.0218 | State Street Communication Services Sel... | Communication Services   | FLAT |
| 9  | ESPO | 0.0216 | VanEck Video Gaming and eSports ETF        | Consumer Discretionary   | FLAT |
| 17 | FPXI | 0.0211 | First Trust International Equity Opport... | International IPO Equity | FLAT |
| 40 | EIS  | 0.0198 | iShares MSCI Israel ETF                    | Israel Equity            | FLAT |
| 41 | BLCN | 0.0192 | Siren NexGen Economy ETF                   | Technology               | FLAT |

## Cluster 8 – Emerging Markets Equity

**Members:** 25   **Var explained:** 0.872   **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-04-10 – filter **PASS** (252d at date = +57.0%)
- Latest sell\_cross: 2026-07-07
- Current 252d return: +30.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-04-10)



### Category breakdown

| category | n_members | abs_weight |
|----------|-----------|------------|
|----------|-----------|------------|

|                         |    |        |
|-------------------------|----|--------|
| Emerging Markets Equity | 19 | 0.7727 |
| Asia Equity             | 2  | 0.0814 |
| Asia ex-Japan Equity    | 1  | 0.0418 |
| Taiwan Equity           | 1  | 0.0362 |
| South Korea Equity      | 1  | 0.0341 |
| Global Dividend Equity  | 1  | 0.0337 |

**Replication basket** – IEMG (cor=0.99) basket lead at lag 1: -0.037

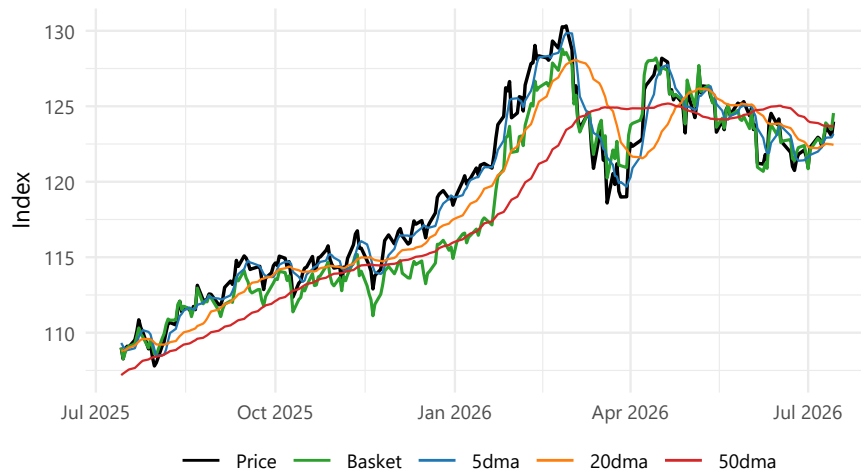
| ticker | rep_wt | name                                   | category                 | etf_signal |
|--------|--------|----------------------------------------|--------------------------|------------|
| IEMG   | 0.92   | iShares Core MSCI Emerging Markets ETF | Emerging Markets Equi... | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                | etf_signal |
|----|--------|--------|--------------------------------------------|-------------------------|------------|
| 21 | IEMG   | 0.0425 | iShares Core MSCI Emerging Markets ETF     | Emerging Markets Equity | FLAT       |
| 2  | EEM    | 0.0424 | iShares MSCI Emerging Index Fund           | Emerging Markets Equity | FLAT       |
| 3  | VWO    | 0.0422 | Vanguard FTSE Emerging Markets ETF         | Emerging Markets Equity | LONG       |
| 16 | GEM    | 0.0422 | Goldman Sachs ActiveBeta Emerging Marke... | Emerging Markets Equity | FLAT       |
| 20 | ESGE   | 0.0422 | iShares ESG Aware MSCI EM ETF              | Emerging Markets Equity | FLAT       |
| 23 | SPEM   | 0.0422 | State Street SPDR Portfolio Emerging Ma... | Emerging Markets Equity | FLAT       |
| 22 | SCHE   | 0.0421 | Schwab Emerging Markets Equity ETF         | Emerging Markets Equity | LONG       |
| 18 | XSOE   | 0.0420 | WisdomTree Emerging Markets Ex-State Ow... | Emerging Markets Equity | FLAT       |
| 15 | EMGF   | 0.0419 | iShares Emerging Markets Equity Factor ... | Emerging Markets Equity | FLAT       |
| 5  | AAXJ   | 0.0418 | iShares MSCI All Country Asia ex Japan ... | Asia ex-Japan Equity    | FLAT       |
| 11 | EEMA   | 0.0414 | iShares MSCI Emerging Markets Asia ETF     | Emerging Markets Equity | FLAT       |
| 14 | GMF    | 0.0412 | State Street SPDR S&P Emerging Asia Pac... | Asia Equity             | FLAT       |
| 12 | PXH    | 0.0409 | Invesco RAFI Emerging Markets ETF          | Emerging Markets Equity | FLAT       |
| 17 | FNDE   | 0.0409 | Schwab Fundamental Emerging Markets Equ... | Emerging Markets Equity | FLAT       |
| 19 | EEMV   | 0.0405 | iShares MSCI Emerging Markets Min Vol F... | Emerging Markets Equity | FLAT       |
| 10 | AIA    | 0.0403 | iShares Asia 50 ETF                        | Asia Equity             | LONG       |
| 8  | DGS    | 0.0402 | WisdomTree Emerging Market SmallCap Fund   | Emerging Markets Equity | FLAT       |
| 7  | DEM    | 0.0397 | WisdomTree Emerging Markets High Divide... | Emerging Markets Equity | FLAT       |
| 6  | EWX    | 0.0392 | State Street SPDR S&P Emerging Markets ... | Emerging Markets Equity | FLAT       |
| 4  | FEM    | 0.0388 | First Trust Emerging Markets AlphaDEX F... | Emerging Markets Equity | FLAT       |
| 13 | DVYE   | 0.0364 | iShares Emerging Markets Dividend Index... | Emerging Markets Equity | FLAT       |
| 9  | EWT    | 0.0362 | iShares MSCI Taiwan ETF                    | Taiwan Equity           | FLAT       |
| 25 | FEMS   | 0.0349 | First Trust Emerging Markets Small Cap ... | Emerging Markets Equity | FLAT       |
| 1  | EWY    | 0.0341 | iShares MSCI South Korea ETF               | South Korea Equity      | FLAT       |
| 24 | SDIV   | 0.0337 | Global X SuperDividend ETF                 | Global Dividend Equity  | FLAT       |

**Cluster 9 – Asia-Pacific Equity****Members:** 24   **Var explained:** 0.680   **Signal:** FLAT**Signal logic:**

- Latest buy\_cross: 2026-05-22 – filter **PASS** (252d at date = +17.3%)
- Latest sell\_cross: 2026-06-23
- Current 252d return: +13.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-23 more recent than last effective buy 2026-05-22)

**Category breakdown**

| category                      | n_members | abs_weight |
|-------------------------------|-----------|------------|
| Asia-Pacific Equity           | 2         | 0.0944     |
| International Dividend Equity | 2         | 0.0919     |
| International Equity          | 2         | 0.0919     |
| Global Equity                 | 2         | 0.0881     |
| Real Estate                   | 2         | 0.0881     |
| Emerging Markets Fixed Income | 2         | 0.0804     |
| Currency                      | 2         | 0.0778     |
| Australia Equity              | 1         | 0.0467     |
| Canada Equity                 | 1         | 0.0443     |
| Switzerland Equity            | 1         | 0.0413     |
| Singapore Equity              | 1         | 0.0399     |

|                      |   |        |
|----------------------|---|--------|
| South Africa Equity  | 1 | 0.0395 |
| Latin America Equity | 1 | 0.0379 |
| New Zealand Equity   | 1 | 0.0379 |
| Mexico Equity        | 1 | 0.0359 |
| Brazil Equity        | 1 | 0.0321 |
| Chile Equity         | 1 | 0.0319 |

**Replication basket** – EWZ+IGF+BBAX (cor=0.97) basket lead at lag 1: -0.003

| ticker | rep_wt | name                                     | category            | etf_signal |
|--------|--------|------------------------------------------|---------------------|------------|
| EWZ    | 0.091  | iShares MSCI Brazil ETF                  | Brazil Equity       | FLAT       |
| IGF    | 0.281  | iShares Global Infrastructure ETF        | Global Equity       | LONG       |
| BBAX   | 0.464  | JPMorgan BetaBuilders Developed Asia ... | Asia-Pacific Equity | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 8  | EPP    | 0.0472 | iShares MSCI Pacific Ex-Japan Index Fund   | Asia-Pacific Equity         | FLAT       |
| 24 | BBAX   | 0.0472 | JPMorgan BetaBuilders Developed Asia Pa... | Asia-Pacific Equity         | LONG       |
| 12 | IDLV   | 0.0469 | Invesco S&P International Developed Low... | International Equity        | LONG       |
| 4  | EWA    | 0.0467 | iShares MSCI Australia Index Fund          | Australia Equity            | FLAT       |
| 11 | DWX    | 0.0463 | State Street SPDR S&P International Div... | International Dividend E... | LONG       |
| 3  | PID    | 0.0457 | Invesco International Dividend Achiever... | International Dividend E... | FLAT       |
| 13 | EFAV   | 0.0450 | iShares MSCI EAFE Min Vol Factor ETF       | International Equity        | FLAT       |
| 9  | NFRA   | 0.0446 | FlexShares STOXX Global Broad Infrastru... | Global Equity               | FLAT       |
| 14 | VNQI   | 0.0446 | Vanguard Global ex-U.S. Real Estate ETF    | Real Estate                 | FLAT       |
| 7  | EWC    | 0.0443 | iShares MSCI Canada Index Fund             | Canada Equity               | LONG       |
| 10 | IGF    | 0.0435 | iShares Global Infrastructure ETF          | Global Equity               | LONG       |
| 2  | RWX    | 0.0434 | State Street SPDR Dow Jones Internation... | Real Estate                 | FLAT       |
| 20 | FXA    | 0.0414 | Invesco CurrencyShares Australian Dolla... | Currency                    | FLAT       |
| 5  | EWL    | 0.0413 | iShares MSCI Switzerland ETF               | Switzerland Equity          | LONG       |
| 15 | EBND   | 0.0403 | State Street SPDR Bloomberg Emerging Ma... | Emerging Markets Fixed I... | FLAT       |
| 16 | EMLC   | 0.0401 | VanEck J. P. Morgan EM Local Currency B... | Emerging Markets Fixed I... | FLAT       |
| 6  | EWS    | 0.0399 | iShares MSCI Singapore ETF                 | Singapore Equity            | LONG       |
| 22 | EZA    | 0.0395 | iShares MSCI South Africa Index Fund       | South Africa Equity         | FLAT       |
| 19 | ILF    | 0.0379 | iShares Latin America 40 ETF               | Latin America Equity        | FLAT       |
| 21 | ENZL   | 0.0379 | iShares MSCI New Zealand ETF               | New Zealand Equity          | LONG       |
| 23 | FXC    | 0.0364 | Invesco CurrencyShares Canadian Dollar ... | Currency                    | FLAT       |
| 17 | EWZ    | 0.0359 | iShares MSCI Mexico ETF                    | Mexico Equity               | FLAT       |
| 1  | EWZ    | 0.0321 | iShares MSCI Brazil ETF                    | Brazil Equity               | FLAT       |

18 ECH 0.0319 iShares MSCI Chile ETF

Chile Equity

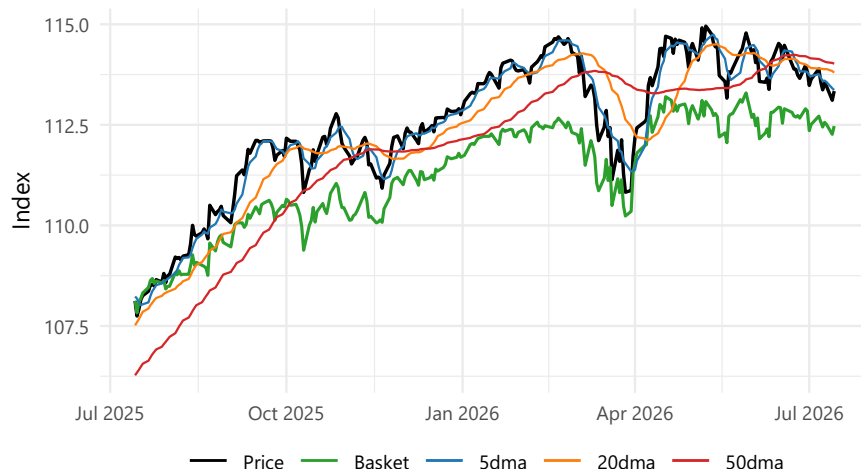
FLAT

## Cluster 10 – High Yield Bond

Members: 21 Var explained: 0.765 Signal: FLAT

### Signal logic:

- Latest buy\_cross: 2026-06-17 – filter **PASS** (252d at date = +7.3%)
- Latest sell\_cross: 2026-07-07
- Current 252d return: +4.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)



### Category breakdown

| category                      | n_members | abs_weight |
|-------------------------------|-----------|------------|
| High Yield Bond               | 10        | 0.5156     |
| Preferred Stock               | 7         | 0.3059     |
| Emerging Markets Fixed Income | 4         | 0.1785     |

**Replication basket** – HYG (cor=0.95) basket lead at lag 1: 0.047

| ticker | rep_wt | name                                      | category        | etf_signal |
|--------|--------|-------------------------------------------|-----------------|------------|
| HYG    | 0.917  | iShares iBoxx \$ High Yield Corporate ... | High Yield Bond | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                        | category                    | etf_signal |
|----|--------|--------|---------------------------------------------|-----------------------------|------------|
| 1  | HYG    | 0.0526 | iShares iBoxx \$ High Yield Corporate Bo... | High Yield Bond             | FLAT       |
| 13 | JNK    | 0.0526 | State Street SPDR Bloomberg High Yield ...  | High Yield Bond             | FLAT       |
| 19 | HYLB   | 0.0524 | Xtrackers USD High Yield Corporate Bond...  | High Yield Bond             | FLAT       |
| 18 | BBHY   | 0.0523 | JPMorgan BetaBuilders USD High Yield Co...  | High Yield Bond             | FLAT       |
| 16 | SHYG   | 0.0521 | iShares 0-5 Year High Yield Corporate B...  | High Yield Bond             | FLAT       |
| 14 | SJNK   | 0.0520 | State Street SPDR Bloomberg Short Term ...  | High Yield Bond             | FLAT       |
| 17 | FALN   | 0.0517 | iShares Fallen Angels USD Bond ETF          | High Yield Bond             | FLAT       |
| 15 | ANGL   | 0.0516 | VanEck Fallen Angel High Yield Bond ETF     | High Yield Bond             | FLAT       |
| 6  | HYS    | 0.0511 | PIMCO 0-5 Year High Yield Corporat Bond...  | High Yield Bond             | FLAT       |
| 9  | PFF    | 0.0479 | iShares Preferred and Income Securities...  | Preferred Stock             | FLAT       |
| 3  | HYLS   | 0.0473 | First Trust Tactical High Yield ETF         | High Yield Bond             | FLAT       |
| 2  | EMB    | 0.0468 | iShares J.P. Morgan USD Emerging Market...  | Emerging Markets Fixed I... | FLAT       |
| 21 | VWOB   | 0.0467 | Vanguard Emerging Markets Government Bo...  | Emerging Markets Fixed I... | FLAT       |
| 11 | PFXF   | 0.0460 | VanEck Preferred Securities ex Financia...  | Preferred Stock             | FLAT       |
| 20 | PCY    | 0.0458 | Invesco Emerging Markets Sovereign Debt...  | Emerging Markets Fixed I... | FLAT       |
| 7  | PGX    | 0.0446 | Invesco Preferred ETF                       | Preferred Stock             | FLAT       |
| 10 | PSK    | 0.0441 | State Street SPDR ICE Preferred Securit...  | Preferred Stock             | FLAT       |
| 4  | FPE    | 0.0433 | First Trust Preferred Securities and In...  | Preferred Stock             | FLAT       |
| 5  | PGF    | 0.0431 | Invesco Financial Preferred ETF             | Preferred Stock             | FLAT       |
| 12 | HYEM   | 0.0392 | VanEck Emerging Markets High Yield Bond...  | Emerging Markets Fixed I... | FLAT       |
| 8  | VRP    | 0.0368 | Invesco Variable Rate Preferred ETF         | Preferred Stock             | LONG       |

## Cluster 11 – Healthcare

**Members:** 17 **Var explained:** 0.760 **Signal:** LONG since 2026-05-06 (46 days, 9.8%)

### Signal logic:

- Latest buy\_cross: 2026-05-06 – filter **PASS** (252d at date = +13.8%)
- Latest sell\_cross: 2026-04-29
- Current 252d return: +24.7% – filter **ALLOWS** new entries today

- Decision: **LONG** (effective buy at 2026-05-06 > last sell at 2026-04-29)



### Category breakdown

| category                  | n_members | abs_weight |
|---------------------------|-----------|------------|
| Healthcare                | 16        | 0.9418     |
| U.S. Small Cap Healthcare | 1         | 0.0582     |

**Replication basket** – VHT (cor=0.93) basket lead at lag 1: -0.002

| ticker | rep_wt | name                     | category   | etf_signal |
|--------|--------|--------------------------|------------|------------|
| VHT    | 1.094  | Vanguard Health Care ETF | Healthcare | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category   | etf_signal |
|----|--------|--------|--------------------------------------------|------------|------------|
| 15 | VHT    | 0.0654 | Vanguard Health Care ETF                   | Healthcare | LONG       |
| 16 | FHLC   | 0.0653 | Fidelity MSCI Health Care Index ETF        | Healthcare | LONG       |
| 9  | FXH    | 0.0647 | First Trust Health Care AlphaDEX           | Healthcare | LONG       |
| 14 | RSPH   | 0.0640 | Invesco S&P 500 Equal Weight Health Car... | Healthcare | LONG       |

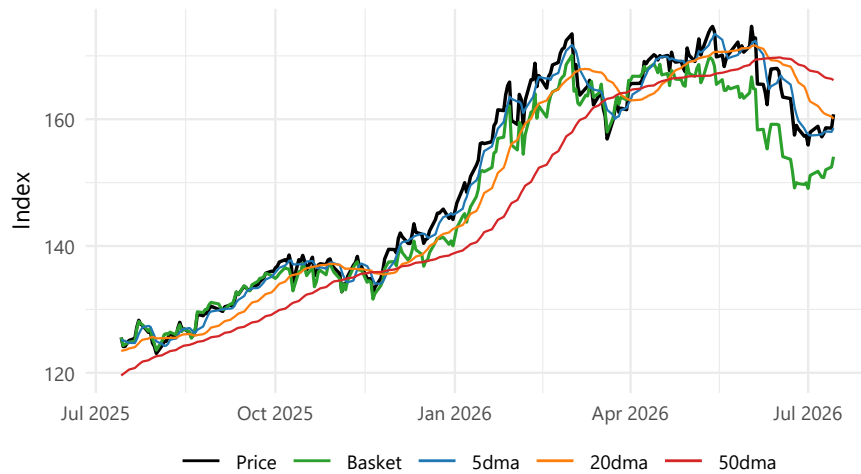
|    |      |        |                                            |                           |      |
|----|------|--------|--------------------------------------------|---------------------------|------|
| 13 | IYH  | 0.0638 | iShares U.S. Healthcare ETF                | Healthcare                | LONG |
| 10 | IXJ  | 0.0623 | iShares Global Healthcare ETF              | Healthcare                | FLAT |
| 1  | XLV  | 0.0622 | State Street Health Care Select Sector ... | Healthcare                | LONG |
| 3  | IBB  | 0.0604 | iShares Biotechnology ETF                  | Healthcare                | LONG |
| 6  | FBT  | 0.0596 | First Trust Amex Biotech Index Fund        | Healthcare                | FLAT |
| 8  | PSCH | 0.0582 | Invesco S&P SmallCap Health Care ETF       | U.S. Small Cap Healthcare | LONG |
| 7  | XHE  | 0.0573 | State Street SPDR S&P Health Care Equip... | Healthcare                | LONG |
| 12 | IHI  | 0.0570 | iShares U.S. Medical Devices ETF           | Healthcare                | FLAT |
| 17 | PPH  | 0.0544 | VanEck Pharmaceutical ETF                  | Healthcare                | LONG |
| 5  | PTH  | 0.0541 | Invesco Dorsey Wright Healthcare Moment... | Healthcare                | LONG |
| 4  | XBI  | 0.0533 | State Street SPDR S&P Biotech ETF          | Healthcare                | LONG |
| 2  | ARKG | 0.0507 | ARK Genomic Revolution ETF                 | Healthcare                | FLAT |
| 11 | IHF  | 0.0472 | iShares U.S. Health Care Providers ETF     | Healthcare                | LONG |

## Cluster 12 – Materials

**Members:** 17   **Var explained:** 0.688   **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-04-07 – filter **PASS** (252d at date = +48.9%)
- Latest sell\_cross: 2026-06-09
- Current 252d return: +27.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-09 more recent than last effective buy 2026-04-07)



### Category breakdown

| category                    | n_members | abs_weight |
|-----------------------------|-----------|------------|
| Materials                   | 9         | 0.5588     |
| MLP / Energy Infrastructure | 4         | 0.2181     |
| Energy                      | 1         | 0.0644     |
| Consumer Staples            | 1         | 0.0598     |
| Norway Equity               | 1         | 0.0555     |
| Industrial Metals           | 1         | 0.0433     |

**Replication basket** – AMLP+URA+GUNR (cor=0.98) basket lead at lag 1: 0.046

| ticker | rep_wt | name                                     | category                 | etf_signal |
|--------|--------|------------------------------------------|--------------------------|------------|
| AMLP   | 0.140  | Alerian MLP ETF                          | MLP / Energy Infrastr... | LONG       |
| URA    | 0.094  | Global X Uranium ETF                     | Materials                | FLAT       |
| GUNR   | 0.776  | FlexShares Global Upstream Natural Re... | Materials                | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

| ticker | PC_wt | name | category | etf_signal |
|--------|-------|------|----------|------------|
|--------|-------|------|----------|------------|

|    |       |        |                                            |                             |      |
|----|-------|--------|--------------------------------------------|-----------------------------|------|
| 11 | GNR   | 0.0691 | State Street SPDR S&P Global Natural Re... | Materials                   | FLAT |
| 10 | GUNR  | 0.0688 | FlexShares Global Upstream Natural Reso... | Materials                   | FLAT |
| 17 | NANR  | 0.0668 | State Street SPDR S&P North American Na... | Materials                   | FLAT |
| 16 | IGE   | 0.0644 | iShares North American Natural Resource... | Energy                      | FLAT |
| 9  | MXI   | 0.0634 | iShares Global Materials ETF               | Materials                   | FLAT |
| 12 | PICK  | 0.0631 | iShares MSCI Global Select Metals & Min... | Materials                   | FLAT |
| 3  | XME   | 0.0614 | State Street SPDR S&P Metals & Mining E... | Materials                   | FLAT |
| 13 | SLX   | 0.0607 | VanEck Steel ETF                           | Materials                   | FLAT |
| 5  | MOO   | 0.0598 | VanEck Agribusiness ETF                    | Consumer Staples            | LONG |
| 2  | COPX  | 0.0589 | Global X Copper Miners ETF                 | Materials                   | FLAT |
| 8  | MLPX  | 0.0566 | Global X MLP & Energy Infrastructure ETF   | MLP / Energy Infrastruct... | LONG |
| 14 | NORW  | 0.0555 | Global X MSCI Norway ETF                   | Norway Equity               | FLAT |
| 4  | EMLP  | 0.0551 | First Trust North American Energy Infra... | MLP / Energy Infrastruct... | LONG |
| 6  | AMLPA | 0.0539 | Alerian MLP ETF                            | MLP / Energy Infrastruct... | LONG |
| 7  | MLPA  | 0.0526 | Global X MLP ETF                           | MLP / Energy Infrastruct... | LONG |
| 1  | URA   | 0.0466 | Global X Uranium ETF                       | Materials                   | FLAT |
| 15 | DBB   | 0.0433 | Invesco DB Base Metals Fund                | Industrial Metals           | FLAT |

### Cluster 13 – China Equity

**Members:** 15   **Var explained:** 0.798   **Signal:** FLAT

#### Signal logic:

- Latest buy\_cross: 2026-04-15 – filter **PASS** (252d at date = +24.3%)
- Latest sell\_cross: 2026-05-20
- Current 252d return: +5.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-04-15)



### Category breakdown

| category                | n_members | abs_weight |
|-------------------------|-----------|------------|
| China Equity            | 13        | 0.8765     |
| Emerging Markets Equity | 1         | 0.0643     |
| Hong Kong Equity        | 1         | 0.0592     |

**Replication basket** – KWEB (cor=0.92) basket lead at lag 1: 0.011

| ticker | rep_wt | name                               | category     | etf_signal |
|--------|--------|------------------------------------|--------------|------------|
| KWEB   | 0.563  | KraneShares CSI China Internet ETF | China Equity | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category     | etf_signal |
|----|--------|--------|--------------------------------------------|--------------|------------|
| 11 | CXSE   | 0.0739 | WisdomTree China ex-State-Owned Enterpr... | China Equity | FLAT       |
| 9  | GXC    | 0.0737 | State Street SPDR S&P China ETF            | China Equity | FLAT       |
| 1  | MCHI   | 0.0731 | iShares MSCI China ETF                     | China Equity | FLAT       |
| 2  | FXI    | 0.0712 | iShares China Large-Cap ETF                | China Equity | FLAT       |
| 5  | CQQQ   | 0.0712 | Invesco China Technology ETF               | China Equity | FLAT       |

|    |      |        |                                            |                         |      |
|----|------|--------|--------------------------------------------|-------------------------|------|
| 7  | CHIQ | 0.0705 | Global X MSCI China Consumer Discretion... | China Equity            | FLAT |
| 4  | KWEB | 0.0680 | KraneShares CSI China Internet ETF         | China Equity            | FLAT |
| 6  | ASHR | 0.0668 | Xtrackers Harvest CSI 300 China A-Share... | China Equity            | FLAT |
| 8  | CNYA | 0.0660 | iShares MSCI China A ETF                   | China Equity            | FLAT |
| 13 | KGRN | 0.0653 | KraneShares MSCI China Clean Technology... | China Equity            | FLAT |
| 3  | EMQQ | 0.0643 | EMQQ The Emerging Markets Internet ETF     | Emerging Markets Equity | FLAT |
| 10 | KBA  | 0.0596 | KraneShares Boseri MSCI China A 50 Conn... | China Equity            | FLAT |
| 12 | EWI  | 0.0592 | iShares MSCI Hong Kong Index Fund          | Hong Kong Equity        | FLAT |
| 14 | CNXT | 0.0592 | VanEck ChiNext Innovators ETF              | China Equity            | FLAT |
| 15 | KURE | 0.0582 | KraneShares MSCI All China Health Care ... | China Equity            | LONG |

## Cluster 14 – Financials

**Members:** 14   **Var explained:** 0.874   **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-04-13 – filter **PASS** (252d at date = +19.9%)
- Latest sell\_cross: 2026-05-11
- Current 252d return: +11.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-11 more recent than last effective buy 2026-04-13)



**Category breakdown**

| category      | n_members | abs_weight |
|---------------|-----------|------------|
| Financials    | 13        | 0.9382     |
| Global Equity | 1         | 0.0618     |

**Replication basket** – XLF (cor=0.96)    basket lead at lag 1: 0.012

| ticker | rep_wt | name                                     | category   | etf_signal |
|--------|--------|------------------------------------------|------------|------------|
| XLF    | 1.095  | State Street Financial Select Sector ... | Financials | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category      | etf_signal |
|----|--------|--------|--------------------------------------------|---------------|------------|
| 10 | VFH    | 0.0756 | Vanguard Financials ETF                    | Financials    | FLAT       |
| 11 | FNCL   | 0.0755 | Fidelity MSCI Financials Index ETF         | Financials    | LONG       |
| 7  | IYF    | 0.0753 | iShares U.S. Financial ETF                 | Financials    | FLAT       |
| 5  | FXO    | 0.0752 | First Trust Financials AlphaDEX            | Financials    | LONG       |
| 8  | IYG    | 0.0748 | iShares U.S. Financial Services ETF        | Financials    | FLAT       |
| 1  | XLF    | 0.0741 | State Street Financial Select Sector SP... | Financials    | LONG       |
| 9  | KBWB   | 0.0733 | Invesco KBW Bank ETF                       | Financials    | FLAT       |
| 3  | KBE    | 0.0712 | State Street SPDR S&P Bank ETF             | Financials    | FLAT       |
| 14 | KCE    | 0.0707 | State Street SPDR S&P Capital Markets E... | Financials    | LONG       |
| 6  | IAT    | 0.0706 | iShares U.S. Regional Banks ETF            | Financials    | LONG       |
| 13 | IAI    | 0.0697 | iShares U.S. Broker-Dealers & Securitie... | Financials    | LONG       |
| 2  | KRE    | 0.0687 | State Street SPDR S&P Regional Banking ... | Financials    | FLAT       |
| 4  | KIE    | 0.0634 | State Street SPDR S&P Insurance ETF        | Financials    | LONG       |
| 12 | PSP    | 0.0618 | Invesco Global Listed Private Equity ETF   | Global Equity | FLAT       |

**Cluster 15 – Real Estate**

**Members:** 14    **Var explained:** 0.961    **Signal:** FLAT

**Signal logic:**

- Latest buy\_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell\_cross: 2026-06-03

- Current 252d return: +13.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2026-04-10)



### Category breakdown

| category    | n_members | abs_weight |
|-------------|-----------|------------|
| Real Estate | 14        | 1          |

**Replication basket** – VNQ (cor=0.99) basket lead at lag 1: -0.002

| ticker | rep_wt | name                     | category    | etf_signal |
|--------|--------|--------------------------|-------------|------------|
| VNQ    | 0.97   | Vanguard Real Estate ETF | Real Estate | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                | category    | etf_signal |
|----|--------|--------|-------------------------------------|-------------|------------|
| 8  | VNQ    | 0.0725 | Vanguard Real Estate ETF            | Real Estate | FLAT       |
| 9  | FREL   | 0.0725 | Fidelity MSCI Real Estate Index ETF | Real Estate | LONG       |
| 11 | SCHH   | 0.0725 | Schwab U.S. REIT ETF                | Real Estate | FLAT       |
| 2  | IYR    | 0.0724 | iShares U.S. Real Estate ETF        | Real Estate | LONG       |

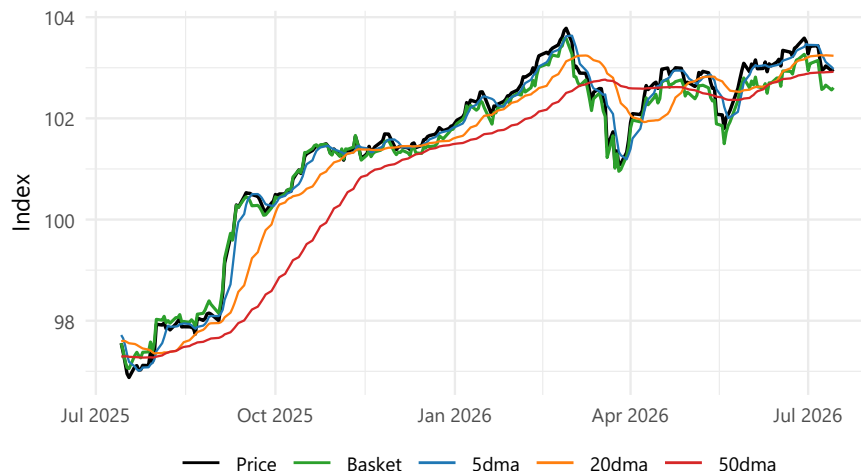
|    |      |        |                                            |             |      |
|----|------|--------|--------------------------------------------|-------------|------|
| 10 | USRT | 0.0724 | iShares Core U.S. REIT ETF                 | Real Estate | FLAT |
| 6  | RWR  | 0.0723 | State Street SPDR Dow Jones REIT ETF       | Real Estate | FLAT |
| 13 | PSR  | 0.0722 | Invesco Active U.S. Real Estate Fund       | Real Estate | FLAT |
| 1  | XLRE | 0.0719 | State Street Real Estate Select Sector ... | Real Estate | LONG |
| 5  | ICF  | 0.0719 | iShares Select U.S. REIT ETF               | Real Estate | FLAT |
| 7  | REET | 0.0716 | iShares Global REIT ETF                    | Real Estate | FLAT |
| 3  | RWO  | 0.0711 | State Street SPDR Dow Jones Global Real... | Real Estate | LONG |
| 4  | REZ  | 0.0697 | iShares Residential and Multisector Rea... | Real Estate | FLAT |
| 14 | NURE | 0.0688 | Nuveen Short-Term REIT ETF                 | Real Estate | LONG |
| 12 | INDS | 0.0682 | Pacer Industrial Real Estate ETF           | Real Estate | LONG |

### Cluster 16 – Municipal Bond

**Members:** 13   **Var explained:** 0.819   **Signal:** FLAT

#### Signal logic:

- Latest buy\_cross: 2026-05-28 – filter **PASS** (252d at date = +5.9%)
- Latest sell\_cross: 2026-07-10
- Current 252d return: +5.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-10 more recent than last effective buy 2026-05-28)



**Category breakdown**

| category       | n_members | abs_weight |
|----------------|-----------|------------|
| Municipal Bond | 13        | 1          |

**Replication basket** – MUB (cor=0.95)    basket lead at lag 1: 0.177

| ticker | rep_wt | name                           | category       | etf_signal |
|--------|--------|--------------------------------|----------------|------------|
| MUB    | 0.904  | iShares National Muni Bond ETF | Municipal Bond | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category       | etf_signal |
|----|--------|--------|--------------------------------------------|----------------|------------|
| 13 | VTEB   | 0.0817 | Vanguard Tax-Exempt Bond ETF               | Municipal Bond | FLAT       |
| 11 | MUB    | 0.0813 | iShares National Muni Bond ETF             | Municipal Bond | FLAT       |
| 6  | CMF    | 0.0802 | iShares California Muni Bond ETF           | Municipal Bond | FLAT       |
| 9  | TFI    | 0.0801 | State Street SPDR Nuveen ICE Municipal ... | Municipal Bond | FLAT       |
| 8  | ITM    | 0.0800 | VanEck Intermediate Muni ETF               | Municipal Bond | FLAT       |
| 1  | FMB    | 0.0794 | First Trust Managed Municipal ETF          | Municipal Bond | FLAT       |
| 7  | NYF    | 0.0793 | iShares New York Muni Bond ETF             | Municipal Bond | FLAT       |
| 5  | PZA    | 0.0780 | Invesco National AMT-Free Municipal Bon... | Municipal Bond | FLAT       |
| 4  | MUNI   | 0.0769 | PIMCO Intermediate Municipal Bond Activ... | Municipal Bond | FLAT       |
| 2  | HYD    | 0.0747 | VanEck High Yield Muni ETF                 | Municipal Bond | FLAT       |
| 3  | HYMB   | 0.0721 | State Street SPDR Nuveen ICE High Yield... | Municipal Bond | FLAT       |
| 12 | SUB    | 0.0695 | iShares Short-Term National Muni Bond E... | Municipal Bond | FLAT       |
| 10 | SHM    | 0.0667 | State Street SPDR Nuveen ICE Short Term... | Municipal Bond | FLAT       |

**Cluster 17 – Gold**

**Members:** 12    **Var explained:** 0.862    **Signal:** FLAT

**Signal logic:**

- Latest buy\_cross: 2026-05-12 – filter **PASS** (252d at date = +75.1%)
- Latest sell\_cross: 2026-05-20
- Current 252d return: +23.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-05-12)



### Category breakdown

| category        | n_members | abs_weight |
|-----------------|-----------|------------|
| Materials       | 5         | 0.4157     |
| Gold            | 3         | 0.2502     |
| Precious Metals | 2         | 0.1724     |
| Silver          | 2         | 0.1617     |

**Replication basket** – GLD+IAU+SLV (cor=0.95)    basket lead at lag 1: -0.011

| ticker | rep_wt | name                      | category | etf_signal |
|--------|--------|---------------------------|----------|------------|
| GLD    | 0.010  | SPDR Gold Shares          | Gold     | FLAT       |
| IAU    | 0.761  | iShares Gold Trust Shares | Gold     | FLAT       |
| SLV    | 0.412  | iShares Silver Trust      | Silver   | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category        | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------|------------|
| 11 | DBP    | 0.0865 | Invesco DB Precious Metals Fund            | Precious Metals | FLAT       |
| 7  | GLTR   | 0.0858 | abrdn Physical Precious Metals Basket S... | Precious Metals | FLAT       |

|    |      |        |                                            |           |      |
|----|------|--------|--------------------------------------------|-----------|------|
| 2  | GDXJ | 0.0841 | VanEck Junior Gold Miners ETF              | Materials | FLAT |
| 1  | GDX  | 0.0840 | VanEck Gold Miners ETF                     | Materials | FLAT |
| 3  | GLD  | 0.0834 | SPDR Gold Shares                           | Gold      | FLAT |
| 9  | IAU  | 0.0834 | iShares Gold Trust Shares                  | Gold      | FLAT |
| 10 | SGOL | 0.0833 | abrdn Physical Gold Shares ETF             | Gold      | FLAT |
| 12 | GOAU | 0.0831 | US Global GO Gold and Precious Metal Mi... | Materials | FLAT |
| 6  | SIL  | 0.0828 | Global X Silver Miners ETF                 | Materials | FLAT |
| 5  | SILJ | 0.0818 | Amplify Junior Silver Miners ETF           | Materials | FLAT |
| 4  | SLV  | 0.0810 | iShares Silver Trust                       | Silver    | LONG |
| 8  | SIVR | 0.0808 | abrdn Physical Silver Shares ETF           | Silver    | LONG |

### Cluster 18 – Consumer Staples

**Members:** 11   **Var explained:** 0.745   **Signal:** LONG since 2026-06-26 (11 days, -0.9%)

#### Signal logic:

- Latest buy\_cross: 2026-06-26 – filter **PASS** (252d at date = +9.9%)
- Latest sell\_cross: 2026-05-29
- Current 252d return: +7.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-26 > last sell at 2026-05-29)



**Category breakdown**

| category                        | n_members | abs_weight |
|---------------------------------|-----------|------------|
| Consumer Staples                | 6         | 0.5732     |
| Utilities                       | 4         | 0.3520     |
| U.S. Small Cap Consumer Staples | 1         | 0.0748     |

**Replication basket** – XLU+XLP (cor=0.99) basket lead at lag 1: -0.015

| ticker | rep_wt | name                                     | category         | etf_signal |
|--------|--------|------------------------------------------|------------------|------------|
| XLU    | 0.364  | State Street Utilities Select Sector ... | Utilities        | LONG       |
| XLP    | 0.587  | State Street Consumer Staples Select ... | Consumer Staples | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 5  | VDC    | 0.0983 | Vanguard Consumer Staples ETF              | Consumer Staples            | LONG       |
| 6  | FSTA   | 0.0983 | Fidelity MSCI Consumer Staples Index ETF   | Consumer Staples            | LONG       |
| 1  | XLP    | 0.0975 | State Street Consumer Staples Select Se... | Consumer Staples            | LONG       |
| 4  | IYK    | 0.0949 | iShares U.S. Consumer Staples ETF          | Consumer Staples            | LONG       |
| 3  | KXI    | 0.0946 | iShares Global Consumer Staples ETF        | Consumer Staples            | LONG       |
| 11 | PBJ    | 0.0897 | Invesco Food & Beverage ETF                | Consumer Staples            | LONG       |
| 7  | IDU    | 0.0890 | iShares U.S. Utilities ETF                 | Utilities                   | LONG       |
| 9  | FUTY   | 0.0878 | Fidelity MSCI Utilities Index ETF          | Utilities                   | LONG       |
| 8  | VPU    | 0.0877 | Vanguard Utilities ETF                     | Utilities                   | LONG       |
| 2  | XLU    | 0.0874 | State Street Utilities Select Sector SP... | Utilities                   | LONG       |
| 10 | PSCC   | 0.0748 | Invesco S&P SmallCap Consumer Staples E... | U.S. Small Cap Consumer ... | LONG       |

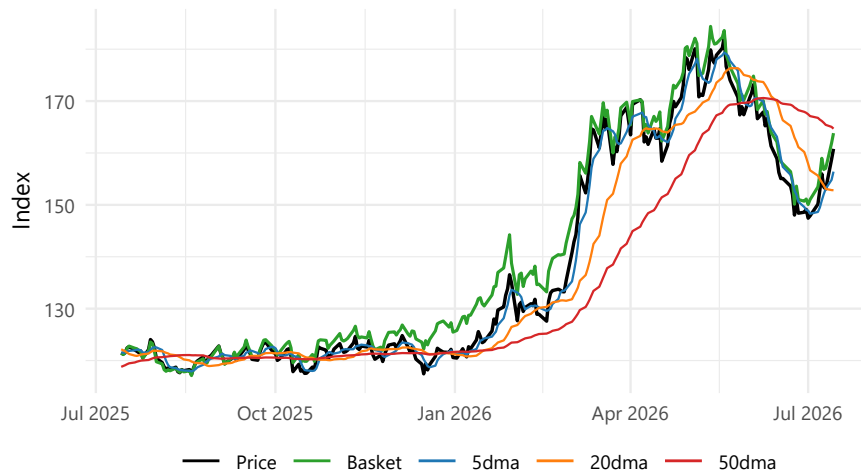
**Cluster 19 – Broad Commodities**

**Members:** 11 **Var explained:** 0.771 **Signal:** LONG since 2026-06-03 (27 days, -7.1%)

**Signal logic:**

- Latest buy\_cross: 2026-06-03 – filter **PASS** (252d at date = +48.1%)
- Latest sell\_cross: 2026-05-27
- Current 252d return: +31.2% – filter **ALLOWS** new entries today

- Decision: **LONG** (effective buy at 2026-06-03 > last sell at 2026-05-27)



### Category breakdown

| category                 | n_members | abs_weight |
|--------------------------|-----------|------------|
| Broad Commodities        | 6         | 0.6175     |
| Oil                      | 2         | 0.1980     |
| Energy Commodities       | 1         | 0.1019     |
| Agricultural Commodities | 1         | 0.0497     |
| Natural Gas              | 1         | 0.0328     |

**Replication basket** – PDBC (cor=0.98) basket lead at lag 1: 0.077

| ticker | rep_wt | name                                     | category          | etf_signal |
|--------|--------|------------------------------------------|-------------------|------------|
| PDBC   | 1.122  | Invesco Optimum Yield Diversified Com... | Broad Commodities | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

| ticker | PC_wt  | name                                     | category          | etf_signal |
|--------|--------|------------------------------------------|-------------------|------------|
| 2 DBC  | 0.1056 | Invesco DB Commodity Index Tracking Fund | Broad Commodities | LONG       |

|    |      |        |                                            |                          |      |
|----|------|--------|--------------------------------------------|--------------------------|------|
| 6  | GSG  | 0.1056 | iShares GSCI Commodity-Indexed Trust Fu... | Broad Commodities        | FLAT |
| 8  | PDBC | 0.1056 | Invesco Optimum Yield Diversified Commo... | Broad Commodities        | LONG |
| 9  | COMT | 0.1052 | iShares GSCI Commodity Dynamic Roll Str... | Broad Commodities        | FLAT |
| 10 | DBE  | 0.1019 | Invesco DB Energy Fund                     | Energy Commodities       | LONG |
| 1  | USO  | 0.0995 | United States Oil Fund                     | Oil                      | LONG |
| 5  | DBO  | 0.0985 | Invesco DB Oil Fund                        | Oil                      | FLAT |
| 7  | DJP  | 0.0983 | iPath Bloomberg Commodity Index Total R... | Broad Commodities        | LONG |
| 4  | FTGC | 0.0972 | First Trust Global Tactical Commodity S... | Broad Commodities        | FLAT |
| 3  | DBA  | 0.0497 | Invesco DB Agriculture Fund                | Agricultural Commodities | LONG |
| 11 | UNG  | 0.0328 | United States Natural Gas Fund LP          | Natural Gas              | FLAT |

## Cluster 20 – India Equity

**Members:** 10   **Var explained:** 0.405   **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-05-06 – filter **PASS** (252d at date = +4.7%)
- Latest sell\_cross: 2026-07-13
- Current 252d return: -14.4% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-13 more recent than last effective buy 2026-05-06)



**Category breakdown**

| category             | n_members | abs_weight |
|----------------------|-----------|------------|
| India Equity         | 3         | 0.4679     |
| Malaysia Equity      | 1         | 0.1124     |
| Indonesia Equity     | 1         | 0.1040     |
| Philippines Equity   | 1         | 0.1011     |
| Vietnam Equity       | 1         | 0.0748     |
| Turkey Equity        | 1         | 0.0613     |
| U.S. Dividend Equity | 1         | 0.0548     |
| Platinum             | 1         | 0.0237     |

**Replication basket** – NOBL+INDA+PPLT (cor=0.91) basket lead at lag 1: -0.006

| ticker | rep_wt | name                                     | category             | etf_signal |
|--------|--------|------------------------------------------|----------------------|------------|
| NOBL   | 0.080  | ProShares S&P 500 Dividend Aristocrat... | U.S. Dividend Equity | FLAT       |
| INDA   | 0.660  | iShares MSCI India ETF                   | India Equity         | FLAT       |
| PPLT   | 0.029  | abrdn Physical Platinum Shares ETF       | Platinum             | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category             | etf_signal |
|----|--------|--------|--------------------------------------------|----------------------|------------|
| 1  | INDA   | 0.1573 | iShares MSCI India ETF                     | India Equity         | FLAT       |
| 2  | INDY   | 0.1557 | iShares India 50 ETF                       | India Equity         | FLAT       |
| 3  | EPI    | 0.1549 | WisdomTree India Earnings Fund             | India Equity         | FLAT       |
| 8  | EWM    | 0.1124 | iShares MSCI Malaysia Index Fund           | Malaysia Equity      | FLAT       |
| 6  | EIDO   | 0.1040 | iShares MSCI Indonesia ETF                 | Indonesia Equity     | FLAT       |
| 7  | EPHE   | 0.1011 | iShares MSCI Philippines ETF               | Philippines Equity   | FLAT       |
| 10 | VNM    | 0.0748 | VanEck Vietnam ETF                         | Vietnam Equity       | FLAT       |
| 9  | TUR    | 0.0613 | iShares MSCI Turkey ETF                    | Turkey Equity        | FLAT       |
| 4  | NOBL   | 0.0548 | ProShares S&P 500 Dividend Aristocrats ... | U.S. Dividend Equity | FLAT       |
| 5  | PPLT   | 0.0237 | abrdn Physical Platinum Shares ETF         | Platinum             | FLAT       |

**Cluster 21 – Inflation-Protected Fixed Income**

**Members:** 10 **Var explained:** 0.905 **Signal:** FLAT

**Signal logic:**

- Latest buy\_cross: 2026-05-27 – filter **PASS** (252d at date = +4.8%)
- Latest sell\_cross: 2026-07-02
- Current 252d return: +1.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-02 more recent than last effective buy 2026-05-27)



### Category breakdown

| category                         | n_members | abs_weight |
|----------------------------------|-----------|------------|
| Inflation-Protected Fixed Income | 10        | 1          |

**Replication basket** – VTIP+SCHP (cor=0.99) basket lead at lag 1: -0.013

| ticker | rep_wt | name                                     | category                 | etf_signal |
|--------|--------|------------------------------------------|--------------------------|------------|
| VTIP   | 0.269  | Vanguard Short-Term Inflation-Protect... | Inflation-Protected F... | FLAT       |
| SCHP   | 0.760  | Schwab U.S. TIPS ETF                     | Inflation-Protected F... | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|   | ticker | PC_wt  | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 4 | TDTF   | 0.1035 | FlexShares iBoxx 5 Year Target Duration... | Inflation-Protected Fixe... | FLAT       |

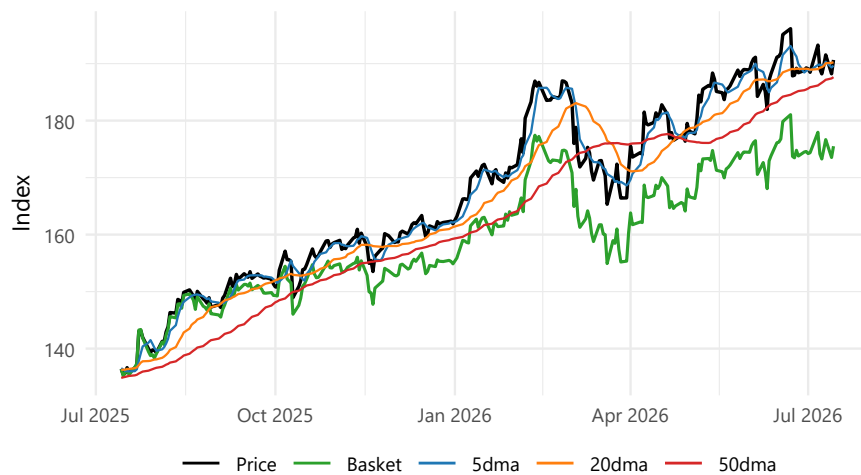
|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 6  | TIPX | 0.1034 | State Street SPDR Bloomberg 1a??10 Year... | Inflation-Protected Fixe... | FLAT |
| 3  | TIP  | 0.1030 | iShares TIPS Bond ETF                      | Inflation-Protected Fixe... | FLAT |
| 8  | SCHP | 0.1023 | Schwab U.S. TIPS ETF                       | Inflation-Protected Fixe... | FLAT |
| 5  | TDDT | 0.1008 | FlexShares iBoxx 3 Year Target Duration... | Inflation-Protected Fixe... | FLAT |
| 7  | SPIP | 0.1000 | State Street SPDR Portfolio TIPS ETF       | Inflation-Protected Fixe... | FLAT |
| 9  | STIP | 0.0998 | iShares 0-5 Year TIPS Bond ETF             | Inflation-Protected Fixe... | FLAT |
| 2  | STPZ | 0.0996 | PIMCO 1-5 Year U.S. TIPS Index Exchange... | Inflation-Protected Fixe... | FLAT |
| 10 | VTIP | 0.0990 | Vanguard Short-Term Inflation-Protected... | Inflation-Protected Fixe... | FLAT |
| 1  | LTPZ | 0.0885 | Pimco 15 Year U.S. TIPS Index Exchange...  | Inflation-Protected Fixe... | FLAT |

## Cluster 22 – Japan Equity

**Members:** 9 **Var explained:** 0.887 **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-04-09 – filter **PASS** (252d at date = +57.2%)
- Latest sell\_cross: 2026-06-11
- Current 252d return: +40.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-11 more recent than last effective buy 2026-04-09)



### Category breakdown

| category            | n_members | abs_weight |
|---------------------|-----------|------------|
| Japan Equity        | 7         | 0.7804     |
| Asia-Pacific Equity | 2         | 0.2196     |

**Replication basket** – EWJ (cor=0.97) basket lead at lag 1: -0.045

| ticker | rep_wt | name                          | category     | etf_signal |
|--------|--------|-------------------------------|--------------|------------|
| EWJ    | 0.91   | iShares MSCI Japan Index Fund | Japan Equity | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

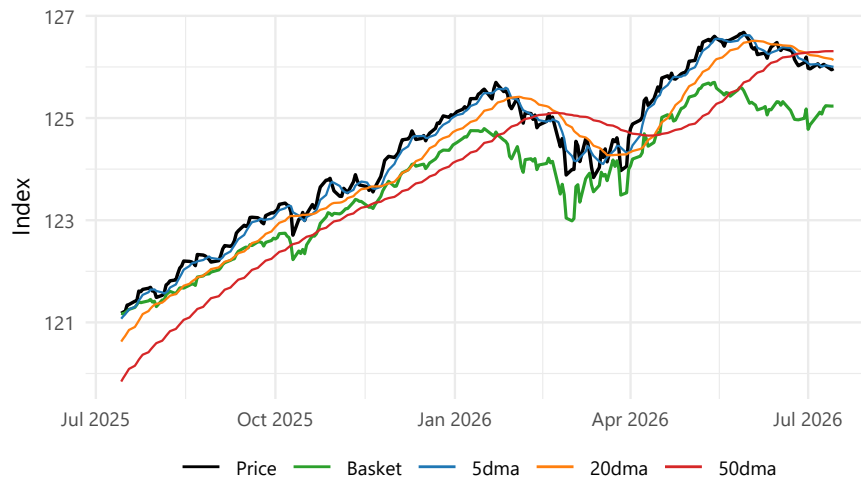
|   | ticker | PC_wt  | name                                   | category            | etf_signal |
|---|--------|--------|----------------------------------------|---------------------|------------|
| 1 | EWJ    | 0.1142 | iShares MSCI Japan Index Fund          | Japan Equity        | FLAT       |
| 8 | BBJP   | 0.1142 | JPMorgan BetaBuilders Japan ETF        | Japan Equity        | FLAT       |
| 9 | HEWJ   | 0.1118 | iShares Currency Hedged MSCI Japan ETF | Japan Equity        | LONG       |
| 6 | DBJP   | 0.1117 | Xtrackers MSCI Japan Hedged Equity ETF | Japan Equity        | LONG       |
| 3 | IPAC   | 0.1114 | iShares Core MSCI Pacific ETF          | Asia-Pacific Equity | LONG       |
| 5 | FLJH   | 0.1109 | Franklin FTSE Japan Hedged ETF         | Japan Equity        | FLAT       |
| 2 | DXJ    | 0.1097 | WisdomTree Japan Hedged Equity Fund    | Japan Equity        | FLAT       |
| 4 | VPL    | 0.1083 | Vanguard FTSE Pacific ETF              | Asia-Pacific Equity | FLAT       |
| 7 | EWJV   | 0.1078 | iShares MSCI Japan Value ETF           | Japan Equity        | LONG       |

### Cluster 23 – Bank Loan / Floating Rate

**Members:** 9 **Var explained:** 0.557 **Signal:** FLAT

#### Signal logic:

- Latest buy\_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell\_cross: 2026-06-23
- Current 252d return: +4.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-23 more recent than last effective buy 2026-04-07)



### Category breakdown

| category                          | n_members | abs_weight |
|-----------------------------------|-----------|------------|
| Bank Loan / Floating Rate         | 3         | 0.3623     |
| Interest Rate Hedged Fixed Income | 3         | 0.3243     |
| Floating Rate Fixed Income        | 3         | 0.3133     |

**Replication basket** – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.047

| ticker | rep_wt | name                                     | category                 | etf_signal |
|--------|--------|------------------------------------------|--------------------------|------------|
| FLOT   | 0.293  | iShares Floating Rate Bond ETF           | Floating Rate Fixed I... | LONG       |
| BKLN   | 0.214  | Invesco Senior Loan ETF                  | Bank Loan / Floating ... | FLAT       |
| SRLN   | 0.302  | State Street Blackstone Senior Loan E... | Bank Loan / Floating ... | FLAT       |
| FLRN   | 0.127  | State Street SPDR Bloomberg Investmen... | Floating Rate Fixed I... | LONG       |
| FLTR   | 0.034  | VanEck IG Floating Rate ETF              | Floating Rate Fixed I... | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

| ticker | PC_wt  | name                                    | category                  | etf_signal |
|--------|--------|-----------------------------------------|---------------------------|------------|
| 2 SRLN | 0.1250 | State Street Blackstone Senior Loan ETF | Bank Loan / Floating Rate | FLAT       |

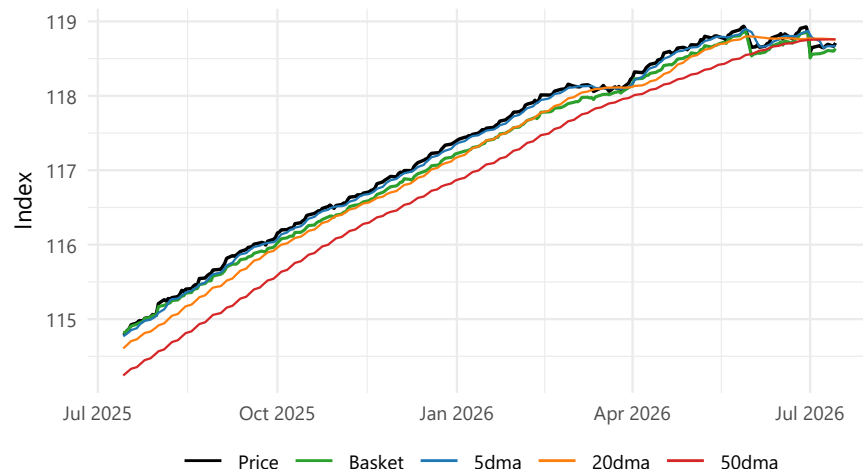
|   |      |        |                                            |                             |      |
|---|------|--------|--------------------------------------------|-----------------------------|------|
| 3 | BKLN | 0.1209 | Invesco Senior Loan ETF                    | Bank Loan / Floating Rate   | FLAT |
| 1 | FTSL | 0.1164 | First Trust Senior Loan Fund               | Bank Loan / Floating Rate   | FLAT |
| 5 | LQDH | 0.1143 | iShares Interest Rate Hedged Corporate ... | Interest Rate Hedged Fix... | FLAT |
| 7 | FLOT | 0.1084 | iShares Floating Rate Bond ETF             | Floating Rate Fixed Inco... | LONG |
| 6 | IGBH | 0.1056 | iShares Interest Rate Hedged Long-Term ... | Interest Rate Hedged Fix... | FLAT |
| 4 | IGHG | 0.1044 | ProShares Investment Grade-Interest Rat... | Interest Rate Hedged Fix... | FLAT |
| 8 | FLRN | 0.1039 | State Street SPDR Bloomberg Investment ... | Floating Rate Fixed Inco... | LONG |
| 9 | FLTR | 0.1011 | VanEck IG Floating Rate ETF                | Floating Rate Fixed Inco... | FLAT |

### Cluster 24 – Cash / Ultra-Short Fixed Income

Members: 9 Var explained: 0.458 Signal: FLAT

#### Signal logic:

- Latest buy\_cross: 2022-10-28 – filter **BLOCKED** (252d at date = -0.5%; needs > 0)
- Latest effective buy: 2022-11-29 – delayed re-arm from blocked cross (252d at fire = +0.0%)
- Latest sell\_cross: 2026-07-07
- Current 252d return: +3.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2022-11-29)



#### Category breakdown

| category                        | n_members | abs_weight |
|---------------------------------|-----------|------------|
| Cash / Ultra-Short Fixed Income | 9         | 1          |

**Replication basket** – SHV+MINT+GBIL+ICSH (cor=0.85) basket lead at lag 1: 0.066

| ticker | rep_wt | name                                     | category                 | etf_signal |
|--------|--------|------------------------------------------|--------------------------|------------|
| SHV    | 0.276  | iShares 0-1 Year Treasury Bond ETF       | Cash / Ultra-Short Fi... | FLAT       |
| MINT   | 0.173  | PIMCO Enhanced Short Maturity Active ... | Cash / Ultra-Short Fi... | FLAT       |
| GBIL   | 0.122  | Goldman Sachs Access Treasury 0-1 Yea... | Cash / Ultra-Short Fi... | FLAT       |
| ICSH   | 0.402  | iShares Ultra Short Duration Bond Act... | Cash / Ultra-Short Fi... | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|   | ticker | PC_wt  | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 7 | SHV    | 0.1306 | iShares 0-1 Year Treasury Bond ETF         | Cash / Ultra-Short Fixed... | FLAT       |
| 6 | ICSH   | 0.1294 | iShares Ultra Short Duration Bond Activ... | Cash / Ultra-Short Fixed... | FLAT       |
| 5 | GSY    | 0.1181 | Invesco Ultra Short Duration ETF           | Cash / Ultra-Short Fixed... | LONG       |
| 2 | MINT   | 0.1126 | PIMCO Enhanced Short Maturity Active Ex... | Cash / Ultra-Short Fixed... | FLAT       |
| 3 | FTSM   | 0.1125 | First Trust Enhanced Short Maturity ETF    | Cash / Ultra-Short Fixed... | FLAT       |
| 4 | NEAR   | 0.1104 | iShares Short Duration Bond Active ETF     | Cash / Ultra-Short Fixed... | FLAT       |
| 8 | BIL    | 0.1015 | State Street SPDR Bloomberg 1-3 Month T... | Cash / Ultra-Short Fixed... | FLAT       |
| 1 | VNLA   | 0.0952 | Janus Henderson Short Duration Income E... | Cash / Ultra-Short Fixed... | FLAT       |
| 9 | GBIL   | 0.0898 | Goldman Sachs Access Treasury 0-1 Year ... | Cash / Ultra-Short Fixed... | FLAT       |

## Cluster 25 – International Fixed Income

**Members:** 6 **Var explained:** 0.742 **Signal:** LONG since 2026-05-29 (30 days, -2.3%)

### Signal logic:

- Latest buy\_cross: 2026-05-29 – filter **PASS** (252d at date = +0.6%)
- Latest sell\_cross: 2026-05-18
- Current 252d return: -3.3% – filter **BLOCKS** new entries today
- Decision: **LONG** (effective buy at 2026-05-29 > last sell at 2026-05-18)



### Category breakdown

| category                   | n_members | abs_weight |
|----------------------------|-----------|------------|
| Currency                   | 4         | 0.6409     |
| International Fixed Income | 2         | 0.3591     |

**Replication basket** – BWX (cor=0.93) basket lead at lag 1: 0.025

| ticker | rep_wt | name                                     | category                 | etf_signal |
|--------|--------|------------------------------------------|--------------------------|------------|
| BWX    | 0.75   | State Street SPDR Bloomberg Internati... | International Fixed I... | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|   | ticker | PC_wt  | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 2 | IGOV   | 0.1801 | iShares International Treasury Bond ETF    | International Fixed Inco... | FLAT       |
| 1 | BWX    | 0.1790 | State Street SPDR Bloomberg Internation... | International Fixed Inco... | FLAT       |
| 4 | FXE    | 0.1702 | Invesco CurrencyShares Euro Currency Tr... | Currency                    | FLAT       |
| 5 | FXF    | 0.1650 | Invesco CurrencyShares Swiss Franc Trust   | Currency                    | FLAT       |
| 3 | FXB    | 0.1609 | Invesco CurrencyShares British Pound St... | Currency                    | FLAT       |
| 6 | FXY    | 0.1448 | Invesco CurrencyShares Japanese Yen Tru... | Currency                    | FLAT       |

**Cluster 26 – Leveraged / Inverse****Members:** 2   **Var explained:** 0.626   **Signal:** FLAT**Signal logic:**

- Latest buy\_cross: 2026-02-05 – filter **BLOCKED** (252d at date = -32.0%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell\_cross: 2026-07-06
- Current 252d return: -35.6% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-06 more recent than last effective buy 2025-04-08)

**Category breakdown**

| category            | n_members | abs_weight |
|---------------------|-----------|------------|
| Currency            | 1         | 0.5        |
| Leveraged / Inverse | 1         | 0.5        |

**Replication basket** – SQQQ+UUP (cor=1.00)   basket lead at lag 1: -0.025

| ticker | rep_wt | name                         | category            | etf_signal |
|--------|--------|------------------------------|---------------------|------------|
| SQQQ   | 0.506  | ProShares UltraPro Short QQQ | Leveraged / Inverse | FLAT       |

|     |       |                                       |          |      |
|-----|-------|---------------------------------------|----------|------|
| UUP | 0.495 | Invesco DB USD Index Bullish Fund ETF | Currency | LONG |
|-----|-------|---------------------------------------|----------|------|

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

| ticker | PC_wt | name                                  | category            | etf_signal |
|--------|-------|---------------------------------------|---------------------|------------|
| SQQQ   | 0.5   | ProShares UltraPro Short QQQ          | Leveraged / Inverse | FLAT       |
| UUP    | 0.5   | Invesco DB USD Index Bullish Fund ETF | Currency            | LONG       |

**Long ETF Positions (full)**

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

| ticker | cluster    | theme                | entry      | days | return | p_sell | nss  |
|--------|------------|----------------------|------------|------|--------|--------|------|
| KIE    | cluster_14 | Financials           | 2026-06-10 | 22   | 9.3%   | 0.00   | 1.00 |
| FXO    | cluster_14 | Financials           | 2026-06-05 | 25   | 7.6%   | 0.00   | 1.00 |
| FNCL   | cluster_14 | Financials           | 2026-04-13 | 63   | 8.8%   | 0.00   | 1.00 |
| XLF    | cluster_14 | Financials           | 2026-04-13 | 63   | 8.7%   | 0.00   | 1.00 |
| IAI    | cluster_14 | Financials           | 2026-07-06 | 6    | 2.2%   | 0.04   | 0.95 |
| IAT    | cluster_14 | Financials           | 2026-05-20 | 36   | 11.0%  | 0.07   | 0.93 |
| KCE    | cluster_14 | Financials           | 2026-07-07 | 5    | 0.6%   | 0.10   | 0.83 |
| IHF    | cluster_11 | Healthcare           | 2026-05-08 | 44   | 14.1%  | 0.00   | 1.00 |
| PSCH   | cluster_11 | Healthcare           | 2026-04-13 | 63   | 23.3%  | 0.00   | 1.00 |
| IBB    | cluster_11 | Healthcare           | 2026-06-17 | 17   | 8.8%   | 0.00   | 1.00 |
| PTH    | cluster_11 | Healthcare           | 2026-06-17 | 17   | 12.3%  | 0.00   | 1.00 |
| XHE    | cluster_11 | Healthcare           | 2026-06-09 | 23   | 2.7%   | 0.00   | 1.00 |
| XBI    | cluster_11 | Healthcare           | 2026-06-16 | 18   | 14.9%  | 0.00   | 1.00 |
| RSPH   | cluster_11 | Healthcare           | 2026-05-21 | 35   | 8.8%   | 0.01   | 0.99 |
| VHT    | cluster_11 | Healthcare           | 2026-05-19 | 37   | 9.0%   | 0.02   | 0.98 |
| FHLC   | cluster_11 | Healthcare           | 2026-05-19 | 37   | 8.9%   | 0.02   | 0.98 |
| FXH    | cluster_11 | Healthcare           | 2026-04-10 | 64   | 8.9%   | 0.03   | 0.97 |
| IYH    | cluster_11 | Healthcare           | 2026-05-20 | 36   | 7.8%   | 0.04   | 0.96 |
| XLV    | cluster_11 | Healthcare           | 2026-05-21 | 35   | 6.8%   | 0.04   | 0.96 |
| PPH    | cluster_11 | Healthcare           | 2026-05-21 | 35   | 2.9%   | 0.21   | 0.79 |
| DLN    | cluster_2  | U.S. Large Cap Value | 2026-04-13 | 63   | 6.3%   | 0.00   | 1.00 |
| DTD    | cluster_2  | U.S. Large Cap Value | 2026-04-13 | 63   | 6.2%   | 0.00   | 1.00 |
| PWV    | cluster_2  | U.S. Large Cap Value | 2026-04-02 | 69   | 12.2%  | 0.00   | 1.00 |
| FTCS   | cluster_2  | U.S. Large Cap Value | 2026-06-26 | 11   | 2.1%   | 0.00   | 1.00 |
| RWL    | cluster_2  | U.S. Large Cap Value | 2026-04-10 | 64   | 10.1%  | 0.00   | 1.00 |
| IWX    | cluster_2  | U.S. Large Cap Value | 2026-04-10 | 64   | 11.7%  | 0.00   | 1.00 |
| USMV   | cluster_2  | U.S. Large Cap Value | 2026-06-29 | 10   | 0.7%   | 0.00   | 1.00 |
| DGRO   | cluster_2  | U.S. Large Cap Value | 2026-04-13 | 63   | 5.9%   | 0.01   | 0.99 |
| IWD    | cluster_2  | U.S. Large Cap Value | 2026-04-10 | 64   | 11.1%  | 0.01   | 0.99 |
| PKW    | cluster_2  | U.S. Large Cap Value | 2026-05-20 | 36   | 4.6%   | 0.01   | 0.99 |
| OUSA   | cluster_2  | U.S. Large Cap Value | 2026-07-01 | 8    | 0.6%   | 0.01   | 0.99 |
| FVD    | cluster_2  | U.S. Large Cap Value | 2026-06-10 | 22   | 2.4%   | 0.01   | 0.99 |
| LGLV   | cluster_2  | U.S. Large Cap Value | 2026-06-26 | 11   | 0.7%   | 0.02   | 0.98 |
| VONV   | cluster_2  | U.S. Large Cap Value | 2026-04-10 | 64   | 11.0%  | 0.02   | 0.98 |
| VIG    | cluster_2  | U.S. Large Cap Value | 2026-04-14 | 62   | 6.0%   | 0.02   | 0.98 |
| SPLV   | cluster_2  | U.S. Large Cap Value | 2026-06-11 | 21   | 2.8%   | 0.03   | 0.97 |
| IMCB   | cluster_2  | U.S. Large Cap Value | 2026-04-10 | 64   | 12.2%  | 0.03   | 0.97 |

|      |            |                       |            |    |       |      |       |
|------|------------|-----------------------|------------|----|-------|------|-------|
| FNDX | cluster_2  | U.S. Large Cap Value  | 2026-04-09 | 65 | 9.2%  | 0.03 | 0.96  |
| SIZE | cluster_2  | U.S. Large Cap Value  | 2026-04-13 | 63 | 7.5%  | 0.05 | 0.95  |
| PRF  | cluster_2  | U.S. Large Cap Value  | 2026-04-09 | 65 | 10.0% | 0.06 | 0.93  |
| SPHD | cluster_2  | U.S. Large Cap Value  | 2026-06-08 | 24 | 4.1%  | 0.07 | 0.93  |
| FTA  | cluster_2  | U.S. Large Cap Value  | 2026-05-21 | 35 | 4.3%  | 0.08 | 0.92  |
| ACWV | cluster_2  | U.S. Large Cap Value  | 2026-07-06 | 6  | -0.4% | 0.01 | 0.92  |
| DIV  | cluster_2  | U.S. Large Cap Value  | 2026-06-29 | 10 | 0.9%  | 0.01 | 0.91  |
| DIA  | cluster_2  | U.S. Large Cap Value  | 2026-04-14 | 62 | 8.1%  | 0.13 | 0.87  |
| LVHD | cluster_2  | U.S. Large Cap Value  | 2026-06-10 | 22 | 2.0%  | 0.13 | 0.87  |
| SDY  | cluster_2  | U.S. Large Cap Value  | 2026-05-20 | 36 | 3.3%  | 0.14 | 0.86  |
| RSP  | cluster_2  | U.S. Large Cap Value  | 2026-04-14 | 62 | 6.9%  | 0.15 | 0.85  |
| HDV  | cluster_2  | U.S. Large Cap Value  | 2026-06-25 | 12 | 0.4%  | 0.12 | 0.79  |
| VO   | cluster_2  | U.S. Large Cap Value  | 2026-04-13 | 63 | 7.8%  | 0.23 | 0.77  |
| AIVL | cluster_2  | U.S. Large Cap Value  | 2026-04-10 | 64 | 8.2%  | 0.35 | 0.65  |
| IWR  | cluster_2  | U.S. Large Cap Value  | 2026-04-10 | 64 | 9.1%  | 0.41 | 0.59  |
| SCHD | cluster_2  | U.S. Large Cap Value  | 2026-07-07 | 5  | -1.0% | 0.11 | 0.51  |
| VTV  | cluster_2  | U.S. Large Cap Value  | 2026-04-13 | 63 | 7.9%  | 0.60 | 0.40  |
| JVAL | cluster_2  | U.S. Large Cap Value  | 2026-04-10 | 64 | 14.1% | 0.75 | 0.22  |
| MGV  | cluster_2  | U.S. Large Cap Value  | 2026-04-13 | 63 | 8.8%  | 0.87 | 0.13  |
| FEX  | cluster_2  | U.S. Large Cap Value  | 2026-04-09 | 65 | 7.8%  | 0.99 | -0.13 |
| GSLC | cluster_5  | U.S. Large Cap Equity | 2026-06-30 | 9  | 0.9%  | 0.01 | 0.96  |
| SPYX | cluster_5  | U.S. Large Cap Equity | 2026-06-30 | 9  | 0.9%  | 0.02 | 0.93  |
| IVV  | cluster_5  | U.S. Large Cap Equity | 2026-06-30 | 9  | 0.8%  | 0.02 | 0.89  |
| SPY  | cluster_5  | U.S. Large Cap Equity | 2026-06-30 | 9  | 0.7%  | 0.03 | 0.87  |
| VOO  | cluster_5  | U.S. Large Cap Equity | 2026-06-30 | 9  | 0.6%  | 0.03 | 0.86  |
| URTH | cluster_5  | U.S. Large Cap Equity | 2026-07-01 | 8  | 0.9%  | 0.03 | 0.84  |
| IWL  | cluster_5  | U.S. Large Cap Equity | 2026-07-01 | 8  | 0.9%  | 0.01 | 0.84  |
| MGC  | cluster_5  | U.S. Large Cap Equity | 2026-07-01 | 8  | 0.8%  | 0.03 | 0.79  |
| DGRW | cluster_5  | U.S. Large Cap Equity | 2026-07-06 | 6  | 0.0%  | 0.02 | 0.79  |
| CATH | cluster_5  | U.S. Large Cap Equity | 2026-07-06 | 6  | 0.4%  | 0.02 | 0.77  |
| OEF  | cluster_5  | U.S. Large Cap Equity | 2026-07-06 | 6  | 0.3%  | 0.00 | 0.75  |
| CRBN | cluster_5  | U.S. Large Cap Equity | 2026-06-30 | 9  | -0.0% | 0.17 | 0.50  |
| IDU  | cluster_18 | Consumer Staples      | 2026-06-25 | 12 | 0.0%  | 0.09 | 0.91  |
| PSCC | cluster_18 | Consumer Staples      | 2026-06-10 | 22 | 5.5%  | 0.16 | 0.84  |
| PBJ  | cluster_18 | Consumer Staples      | 2026-07-13 | 1  | -1.0% | 0.02 | 0.83  |
| XLU  | cluster_18 | Consumer Staples      | 2026-06-25 | 12 | -0.3% | 0.24 | 0.76  |
| FUTY | cluster_18 | Consumer Staples      | 2026-06-25 | 12 | -0.3% | 0.24 | 0.75  |
| IYK  | cluster_18 | Consumer Staples      | 2026-06-09 | 23 | 0.8%  | 0.27 | 0.70  |
| VPU  | cluster_18 | Consumer Staples      | 2026-06-25 | 12 | -0.4% | 0.31 | 0.69  |
| KXI  | cluster_18 | Consumer Staples      | 2026-07-02 | 7  | -1.9% | 0.50 | -0.22 |
| XLP  | cluster_18 | Consumer Staples      | 2026-07-07 | 5  | -1.7% | 0.70 | -0.49 |
| FSTA | cluster_18 | Consumer Staples      | 2026-07-08 | 4  | -1.1% | 0.70 | -0.49 |
| VDC  | cluster_18 | Consumer Staples      | 2026-07-08 | 4  | -1.1% | 0.75 | -0.58 |
| EWS  | cluster_9  | Asia-Pacific Equity   | 2026-06-17 | 17 | 6.9%  | 0.00 | 1.00  |
| ENZL | cluster_9  | Asia-Pacific Equity   | 2026-07-07 | 5  | 2.2%  | 0.00 | 0.99  |

|      |            |                            |            |    |       |      |       |
|------|------------|----------------------------|------------|----|-------|------|-------|
| IDLV | cluster_9  | Asia-Pacific Equity        | 2026-07-10 | 2  | 0.7%  | 0.01 | 0.89  |
| EWC  | cluster_9  | Asia-Pacific Equity        | 2026-07-10 | 2  | 0.9%  | 0.02 | 0.86  |
| BBAX | cluster_9  | Asia-Pacific Equity        | 2026-07-14 | 0  | 0.0%  | 0.02 | 0.78  |
| IGF  | cluster_9  | Asia-Pacific Equity        | 2026-07-09 | 3  | 0.6%  | 0.12 | 0.75  |
| EWL  | cluster_9  | Asia-Pacific Equity        | 2026-06-25 | 12 | -0.6% | 0.22 | 0.71  |
| DWX  | cluster_9  | Asia-Pacific Equity        | 2026-07-09 | 3  | 0.9%  | 0.06 | 0.65  |
| RWO  | cluster_15 | Real Estate                | 2026-06-25 | 12 | 1.3%  | 0.21 | 0.77  |
| INDS | cluster_15 | Real Estate                | 2026-06-08 | 24 | 4.3%  | 0.24 | 0.75  |
| IYR  | cluster_15 | Real Estate                | 2026-06-25 | 12 | 0.5%  | 0.45 | 0.33  |
| FREL | cluster_15 | Real Estate                | 2026-06-25 | 12 | 0.4%  | 0.60 | 0.09  |
| NURE | cluster_15 | Real Estate                | 2026-04-13 | 63 | 12.7% | 0.95 | 0.03  |
| XLRE | cluster_15 | Real Estate                | 2026-06-25 | 12 | -0.2% | 0.81 | -0.56 |
| EWJV | cluster_22 | Japan Equity               | 2026-07-06 | 6  | 0.1%  | 0.01 | 0.96  |
| IPAC | cluster_22 | Japan Equity               | 2026-06-30 | 9  | 1.4%  | 0.32 | 0.42  |
| DBJP | cluster_22 | Japan Equity               | 2026-04-08 | 66 | 7.3%  | 0.60 | 0.35  |
| HEWJ | cluster_22 | Japan Equity               | 2026-04-08 | 66 | 8.1%  | 0.63 | 0.33  |
| SMMV | cluster_1  | U.S. Mid Cap Equity        | 2026-06-25 | 12 | 2.9%  | 0.00 | 1.00  |
| XMLV | cluster_1  | U.S. Mid Cap Equity        | 2026-06-25 | 12 | 1.7%  | 0.01 | 0.99  |
| CGW  | cluster_1  | U.S. Mid Cap Equity        | 2026-06-25 | 12 | 0.7%  | 0.14 | 0.86  |
| REM  | cluster_1  | U.S. Mid Cap Equity        | 2026-07-09 | 3  | 0.9%  | 0.04 | 0.85  |
| PHO  | cluster_1  | U.S. Mid Cap Equity        | 2026-06-26 | 11 | -0.8% | 0.34 | 0.66  |
| REGL | cluster_1  | U.S. Mid Cap Equity        | 2026-06-12 | 20 | 1.5%  | 0.45 | 0.55  |
| SDVY | cluster_1  | U.S. Mid Cap Equity        | 2026-06-08 | 24 | 3.7%  | 0.59 | 0.41  |
| ISMD | cluster_1  | U.S. Mid Cap Equity        | 2026-04-09 | 65 | 16.2% | 0.61 | 0.39  |
| FIW  | cluster_1  | U.S. Mid Cap Equity        | 2026-06-16 | 18 | 1.1%  | 0.76 | 0.24  |
| IJS  | cluster_1  | U.S. Mid Cap Equity        | 2026-04-10 | 64 | 10.2% | 0.77 | 0.22  |
| XRT  | cluster_1  | U.S. Mid Cap Equity        | 2026-05-28 | 31 | 2.5%  | 0.78 | 0.22  |
| FTXR | cluster_1  | U.S. Mid Cap Equity        | 2026-04-17 | 59 | 8.2%  | 0.78 | 0.21  |
| FDIS | cluster_1  | U.S. Mid Cap Equity        | 2026-07-01 | 8  | -1.5% | 0.37 | 0.18  |
| PSCD | cluster_1  | U.S. Mid Cap Equity        | 2026-05-27 | 32 | 7.4%  | 0.84 | 0.16  |
| VCR  | cluster_1  | U.S. Mid Cap Equity        | 2026-07-01 | 8  | -1.6% | 0.39 | 0.16  |
| PEJ  | cluster_1  | U.S. Mid Cap Equity        | 2026-05-26 | 33 | 6.2%  | 0.94 | 0.06  |
| XLY  | cluster_1  | U.S. Mid Cap Equity        | 2026-07-06 | 6  | -1.8% | 0.36 | -0.16 |
| IYC  | cluster_1  | U.S. Mid Cap Equity        | 2026-07-06 | 6  | -1.5% | 0.54 | -0.24 |
| RSPM | cluster_1  | U.S. Mid Cap Equity        | 2026-06-16 | 18 | -4.1% | 0.99 | -0.96 |
| IBUY | cluster_7  | U.S. Extended Market Eq... | 2026-06-15 | 19 | 3.6%  | 0.00 | 1.00  |
| SKYY | cluster_7  | U.S. Extended Market Eq... | 2026-07-01 | 8  | 2.2%  | 0.01 | 0.97  |
| FDN  | cluster_7  | U.S. Extended Market Eq... | 2026-07-07 | 5  | 1.4%  | 0.00 | 0.96  |
| ARKW | cluster_7  | U.S. Extended Market Eq... | 2026-07-02 | 7  | 2.2%  | 0.03 | 0.90  |
| ARKK | cluster_7  | U.S. Extended Market Eq... | 2026-06-29 | 10 | -1.4% | 0.28 | 0.66  |
| IMCG | cluster_7  | U.S. Extended Market Eq... | 2026-04-09 | 65 | 16.5% | 0.60 | 0.40  |
| IWP  | cluster_7  | U.S. Extended Market Eq... | 2026-04-15 | 61 | 4.6%  | 0.62 | 0.27  |
| VOT  | cluster_7  | U.S. Extended Market Eq... | 2026-04-14 | 62 | 8.7%  | 0.78 | 0.15  |
| DBA  | cluster_19 | Broad Commodities          | 2026-07-09 | 3  | -0.3% | 0.00 | 1.00  |
| DJP  | cluster_19 | Broad Commodities          | 2026-06-03 | 27 | -5.2% | 0.00 | 0.01  |

|      |            |                        |            |    |        |      |       |
|------|------------|------------------------|------------|----|--------|------|-------|
| DBC  | cluster_19 | Broad Commodities      | 2026-06-03 | 27 | -5.5%  | 0.00 | 0.00  |
| PDBC | cluster_19 | Broad Commodities      | 2026-06-03 | 27 | -5.4%  | 0.00 | 0.00  |
| DBE  | cluster_19 | Broad Commodities      | 2026-06-03 | 27 | -8.0%  | 0.01 | 0.00  |
| USO  | cluster_19 | Broad Commodities      | 2026-06-03 | 27 | -14.7% | 0.07 | -0.07 |
| KURE | cluster_13 | China Equity           | 2026-07-06 | 6  | 0.5%   | 0.00 | 1.00  |
| UUP  | cluster_26 | Leveraged / Inverse    | 2026-05-18 | 38 | 2.5%   | 0.41 | 0.59  |
| EUFN | cluster_3  | Developed ex-US Equity | 2026-06-12 | 20 | 3.1%   | 0.00 | 1.00  |
| VIGI | cluster_3  | Developed ex-US Equity | 2026-06-30 | 9  | 1.8%   | 0.02 | 0.97  |
| EPOL | cluster_3  | Developed ex-US Equity | 2026-07-08 | 4  | 2.5%   | 0.03 | 0.84  |
| QEFA | cluster_3  | Developed ex-US Equity | 2026-07-06 | 6  | -1.0%  | 0.03 | 0.83  |
| EFV  | cluster_3  | Developed ex-US Equity | 2026-07-08 | 4  | 1.1%   | 0.00 | 0.81  |
| HDEF | cluster_3  | Developed ex-US Equity | 2026-07-09 | 3  | 0.3%   | 0.02 | 0.76  |
| VYMI | cluster_3  | Developed ex-US Equity | 2026-07-08 | 4  | 1.2%   | 0.08 | 0.76  |
| IPKW | cluster_3  | Developed ex-US Equity | 2026-07-10 | 2  | 0.3%   | 0.00 | 0.71  |
| ESGD | cluster_3  | Developed ex-US Equity | 2026-06-30 | 9  | 0.3%   | 0.28 | 0.45  |
| IQLT | cluster_3  | Developed ex-US Equity | 2026-07-01 | 8  | 0.8%   | 0.30 | 0.44  |
| EWU  | cluster_3  | Developed ex-US Equity | 2026-07-07 | 5  | -1.7%  | 0.16 | 0.33  |
| EWO  | cluster_3  | Developed ex-US Equity | 2026-04-10 | 64 | 10.1%  | 0.72 | 0.27  |
| GSIE | cluster_3  | Developed ex-US Equity | 2026-07-02 | 7  | 0.2%   | 0.49 | 0.24  |
| IVLU | cluster_3  | Developed ex-US Equity | 2026-07-07 | 5  | 0.1%   | 0.16 | 0.14  |
| EFA  | cluster_3  | Developed ex-US Equity | 2026-07-01 | 8  | 0.9%   | 0.41 | 0.10  |
| INTF | cluster_3  | Developed ex-US Equity | 2026-07-07 | 5  | 0.2%   | 0.27 | 0.09  |
| IEV  | cluster_3  | Developed ex-US Equity | 2026-07-01 | 8  | 0.5%   | 0.47 | 0.04  |
| IEFA | cluster_3  | Developed ex-US Equity | 2026-07-06 | 6  | -1.4%  | 0.41 | -0.08 |
| VGK  | cluster_3  | Developed ex-US Equity | 2026-07-01 | 8  | 0.6%   | 0.64 | -0.16 |
| IEUR | cluster_3  | Developed ex-US Equity | 2026-07-02 | 7  | -1.3%  | 0.47 | -0.18 |
| MOO  | cluster_12 | Materials              | 2026-07-07 | 5  | 0.7%   | 0.00 | 1.00  |
| EMLP | cluster_12 | Materials              | 2026-07-08 | 4  | 0.5%   | 0.00 | 1.00  |
| MLPX | cluster_12 | Materials              | 2026-07-09 | 3  | 1.3%   | 0.00 | 0.97  |
| AMLP | cluster_12 | Materials              | 2026-07-10 | 2  | 2.1%   | 0.00 | 0.96  |
| MLPA | cluster_12 | Materials              | 2026-07-10 | 2  | 1.8%   | 0.00 | 0.95  |
| SCHG | cluster_6  | U.S. Large Cap Growth  | 2026-07-06 | 6  | 0.1%   | 0.00 | 0.92  |
| QQEW | cluster_6  | U.S. Large Cap Growth  | 2026-04-16 | 60 | 16.5%  | 0.19 | 0.81  |
| IVW  | cluster_6  | U.S. Large Cap Growth  | 2026-07-01 | 8  | 0.7%   | 0.04 | 0.68  |
| VOOG | cluster_6  | U.S. Large Cap Growth  | 2026-07-01 | 8  | 0.8%   | 0.05 | 0.67  |
| SPYG | cluster_6  | U.S. Large Cap Growth  | 2026-07-01 | 8  | 0.8%   | 0.05 | 0.66  |
| IUSG | cluster_6  | U.S. Large Cap Growth  | 2026-07-01 | 8  | 0.6%   | 0.06 | 0.64  |
| MGK  | cluster_6  | U.S. Large Cap Growth  | 2026-07-06 | 6  | 0.3%   | 0.03 | 0.51  |
| VUG  | cluster_6  | U.S. Large Cap Growth  | 2026-07-10 | 2  | -0.5%  | 0.03 | 0.48  |
| ILCG | cluster_6  | U.S. Large Cap Growth  | 2026-06-30 | 9  | -0.8%  | 0.18 | 0.22  |
| ONEQ | cluster_6  | U.S. Large Cap Growth  | 2026-07-10 | 2  | -0.6%  | 0.15 | 0.15  |
| QTUM | cluster_6  | U.S. Large Cap Growth  | 2026-04-10 | 64 | 28.4%  | 1.00 | -0.99 |
| PTF  | cluster_6  | U.S. Large Cap Growth  | 2026-04-07 | 67 | 21.6%  | 1.00 | -1.00 |
| SIVR | cluster_17 | Gold                   | 2026-05-27 | 32 | -21.2% | 0.80 | -0.80 |
| SLV  | cluster_17 | Gold                   | 2026-05-27 | 32 | -21.2% | 0.80 | -0.80 |

|      |            |                            |            |   |       |      |       |
|------|------------|----------------------------|------------|---|-------|------|-------|
| GSY  | cluster_24 | Cash / Ultra-Short Fixe... | 2026-07-07 | 5 | 0.1%  | 0.39 | 0.23  |
| SCHE | cluster_8  | Emerging Markets Equity    | 2026-07-14 | 0 | 0.0%  | 0.49 | 0.04  |
| VWO  | cluster_8  | Emerging Markets Equity    | 2026-07-10 | 2 | -1.4% | 0.63 | -0.30 |
| AIA  | cluster_8  | Emerging Markets Equity    | 2026-07-10 | 2 | -1.9% | 0.91 | -0.82 |
| FLOT | cluster_23 | Bank Loan / Floating Ra... | 2026-07-10 | 2 | -0.0% | 0.70 | -0.45 |
| FLRN | cluster_23 | Bank Loan / Floating Ra... | 2026-07-13 | 1 | -0.0% | 0.73 | -0.45 |
| VRP  | cluster_10 | High Yield Bond            | 2026-07-02 | 7 | -0.2% | 0.40 | 0.38  |
| BSCQ | cluster_4  | Aggregate Fixed Income     | 2026-07-09 | 3 | 0.0%  | 0.29 | 0.30  |

## ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

### Top 20 most bullish (cluster-deduped, abs-mom approved)

| ticker | cluster    | theme                        | p_buy | nss  |
|--------|------------|------------------------------|-------|------|
| EWS    | cluster_9  | Asia-Pacific Equity          | 1.00  | 1.00 |
| KURE   | cluster_13 | China Equity                 | 1.00  | 1.00 |
| KIE    | cluster_14 | Financials                   | 1.00  | 1.00 |
| HACK   | cluster_7  | U.S. Extended Market Equi... | 1.00  | 1.00 |
| DBA    | cluster_19 | Broad Commodities            | 1.00  | 1.00 |
| MOO    | cluster_12 | Materials                    | 1.00  | 1.00 |
| IHF    | cluster_11 | Healthcare                   | 1.00  | 1.00 |
| ILCV   | cluster_2  | U.S. Large Cap Value         | 1.00  | 1.00 |
| SMMV   | cluster_1  | U.S. Mid Cap Equity          | 1.00  | 1.00 |
| QUS    | cluster_5  | U.S. Large Cap Equity        | 1.00  | 1.00 |
| EUFN   | cluster_3  | Developed ex-US Equity       | 1.00  | 1.00 |
| EWJV   | cluster_22 | Japan Equity                 | 0.97  | 0.96 |
| REZ    | cluster_15 | Real Estate                  | 1.00  | 0.95 |
| PTLC   | cluster_6  | U.S. Large Cap Growth        | 0.97  | 0.95 |
| IDU    | cluster_18 | Consumer Staples             | 1.00  | 0.91 |
| VRP    | cluster_10 | High Yield Bond              | 0.78  | 0.38 |
| BSCQ   | cluster_4  | Aggregate Fixed Income       | 0.59  | 0.30 |
| GSY    | cluster_24 | Cash / Ultra-Short Fixed ... | 0.62  | 0.23 |
| BKLN   | cluster_23 | Bank Loan / Floating Rate    | 0.20  | 0.17 |
| SCHE   | cluster_8  | Emerging Markets Equity      | 0.53  | 0.04 |

### Top 20 most bearish (cluster-deduped)

| ticker | cluster    | theme                        | p_sell | nss   |
|--------|------------|------------------------------|--------|-------|
| IGBH   | cluster_23 | Bank Loan / Floating Rate    | 1.00   | -1.00 |
| LTPZ   | cluster_21 | Inflation-Protected Fixed... | 1.00   | -1.00 |
| VCLT   | cluster_4  | Aggregate Fixed Income       | 1.00   | -1.00 |
| XMMO   | cluster_7  | U.S. Extended Market Equi... | 1.00   | -1.00 |
| SUB    | cluster_16 | Municipal Bond               | 1.00   | -1.00 |
| XME    | cluster_12 | Materials                    | 1.00   | -1.00 |

|      |            |                              |      |       |
|------|------------|------------------------------|------|-------|
| UNG  | cluster_19 | Broad Commodities            | 1.00 | -1.00 |
| PTF  | cluster_6  | U.S. Large Cap Growth        | 1.00 | -1.00 |
| HYS  | cluster_10 | High Yield Bond              | 1.00 | -1.00 |
| IYM  | cluster_1  | U.S. Mid Cap Equity          | 1.00 | -0.99 |
| BWX  | cluster_25 | International Fixed Income   | 0.99 | -0.99 |
| EWK  | cluster_3  | Developed ex-US Equity       | 1.00 | -0.99 |
| EWY  | cluster_8  | Emerging Markets Equity      | 1.00 | -0.99 |
| AOM  | cluster_5  | U.S. Large Cap Equity        | 1.00 | -0.99 |
| TUR  | cluster_20 | India Equity                 | 0.97 | -0.97 |
| NEAR | cluster_24 | Cash / Ultra-Short Fixed ... | 0.96 | -0.96 |
| CNYA | cluster_13 | China Equity                 | 0.97 | -0.95 |
| GOAU | cluster_17 | Gold                         | 0.89 | -0.89 |
| VPL  | cluster_22 | Japan Equity                 | 0.96 | -0.86 |
| EWJ  | cluster_9  | Asia-Pacific Equity          | 0.85 | -0.85 |

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## Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

| ticker | name                               | category                 | cluster_id | cluster_size | etf_signal | max_corr_to_others | median_corr_to_others | abs_pc1_loading | compo |
|--------|------------------------------------|--------------------------|------------|--------------|------------|--------------------|-----------------------|-----------------|-------|
| PPLT   | abrdn Physical Platinum Shares ... | Platinum                 | cluster_20 | 10           | FLAT       | 0.305              | 0.095                 | 0.00021         |       |
| UNG    | United States Natural Gas Fund ... | Natural Gas              | cluster_19 | 11           | FLAT       | 0.481              | 0.077                 | 0.00017         |       |
| GBIL   | Goldman Sachs Access Treasury 0... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.741              | 0.038                 | 0.00008         |       |
| FXV    | Invesco CurrencyShares Japanese... | Currency                 | cluster_25 | 6            | FLAT       | 0.730              | 0.094                 | 0.00022         |       |
| VNLA   | Janus Henderson Short Duration ... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.634              | 0.213                 | 0.00036         |       |
| DBA    | Invesco DB Agriculture Fund        | Agricultural Commodit... | cluster_19 | 11           | LONG       | 0.597              | 0.169                 | 0.00040         |       |
| FTSM   | First Trust Enhanced Short Matu... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.742              | 0.125                 | 0.00031         |       |
| ICSH   | iShares Ultra Short Duration Bo... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.742              | 0.155                 | 0.00039         |       |
| MINT   | PIMCO Enhanced Short Maturity A... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.679              | 0.180                 | 0.00046         |       |
| NEAR   | iShares Short Duration Bond Act... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.698              | 0.191                 | 0.00045         |       |
| GSY    | Invesco Ultra Short Duration ETF   | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.730              | 0.180                 | 0.00044         |       |
| TUR    | iShares MSCI Turkey ETF            | Turkey Equity            | cluster_20 | 10           | FLAT       | 0.501              | 0.219                 | 0.00056         |       |
| LDUR   | PIMCO Enhanced Low Duration Act... | Fixed Income             | cluster_4  | 58           | FLAT       | 0.847              | 0.127                 | 0.00036         |       |
| BIL    | State Street SPDR Bloomberg 1-3... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.869              | 0.020                 | 0.00001         |       |
| SHV    | iShares 0-1 Year Treasury Bond ... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.869              | 0.076                 | 0.00021         |       |
| CMBS   | iShares CMBS Bond ETF              | Mortgage-Backed Fixed... | cluster_4  | 58           | FLAT       | 0.876              | 0.160                 | 0.00031         |       |
| FXF    | Invesco CurrencyShares Swiss Fr... | Currency                 | cluster_25 | 6            | FLAT       | 0.820              | 0.204                 | 0.00049         |       |
| LMBS   | First Trust Low Duration Opport... | Mortgage-Backed Fixed... | cluster_4  | 58           | FLAT       | 0.834              | 0.237                 | 0.00047         |       |
| NOBL   | ProShares S&P 500 Dividend Aris... | U.S. Dividend Equity     | cluster_20 | 10           | FLAT       | 0.447              | 0.274                 | 0.00075         |       |
| SHM    | State Street SPDR Nuveen ICE Sh... | Municipal Bond           | cluster_16 | 13           | FLAT       | 0.877              | 0.248                 | 0.00046         |       |
| VNM    | VanEck Vietnam ETF                 | Vietnam Equity           | cluster_20 | 10           | FLAT       | 0.445              | 0.328                 | 0.00083         |       |
| KURE   | KraneShares MSCI All China Heal... | China Equity             | cluster_13 | 15           | LONG       | 0.791              | 0.270                 | 0.00071         |       |
| SUB    | iShares Short-Term National Mun... | Municipal Bond           | cluster_16 | 13           | FLAT       | 0.877              | 0.263                 | 0.00054         |       |
| DBB    | Invesco DB Base Metals Fund        | Industrial Metals        | cluster_12 | 17           | FLAT       | 0.748              | 0.360                 | 0.00085         |       |
| EIDO   | iShares MSCI Indonesia ETF         | Indonesia Equity         | cluster_20 | 10           | FLAT       | 0.500              | 0.351                 | 0.00102         |       |
| FLTR   | VanEck IG Floating Rate ETF        | Floating Rate Fixed I... | cluster_23 | 9            | FLAT       | 0.866              | 0.324                 | 0.00071         |       |
| EPHE   | iShares MSCI Philippines ETF       | Philippines Equity       | cluster_20 | 10           | FLAT       | 0.561              | 0.405                 | 0.00107         |       |
| KBA    | KraneShares Bosera MSCI China A... | China Equity             | cluster_13 | 15           | FLAT       | 0.857              | 0.257                 | 0.00073         |       |
| CNXT   | VanEck ChiNext Innovators ETF      | China Equity             | cluster_13 | 15           | FLAT       | 0.903              | 0.238                 | 0.00064         |       |
| LTPZ   | Pimco 15 Year U.S. TIPS Index E... | Inflation-Protected F... | cluster_21 | 10           | FLAT       | 0.918              | 0.226                 | 0.00057         |       |

## Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

|                          | scope  | parameter                | value |
|--------------------------|--------|--------------------------|-------|
| short_ma                 | screen | short_ma                 | 5     |
| buy_ma                   | screen | buy_ma                   | 50    |
| sell_ma                  | screen | sell_ma                  | 20    |
| vol_window               | screen | vol_window               | 20    |
| horizon                  | screen | horizon                  | 5     |
| imminent_horizon         | screen | imminent_horizon         | 1     |
| cluster_smoothing_window | screen | cluster_smoothing_window | 3     |
| sell_cushion_k           | screen | sell_cushion_k           | 0.5   |
| sell_cushion_sd_window   | screen | sell_cushion_sd_window   | 20    |